
Linear Algebra I-II

Theorems & Lemmas

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Contents

1	Vector Spaces	1
1.1	Vector Spaces and Linear Subspaces	1
1.2	Span of Sets	3
1.3	Bases of Vector Spaces	4
1.3.1	Linear Independence	4
1.3.2	Basis of a Vector Space	5
1.3.3	Example: Bases	6
1.4	Row and Column Spaces	7
1.4.1	Transpose of a Matrix	8
1.5	Sums of Vector Spaces	8
2	Linear Maps	10
2.1	Kernel & Image	11
2.1.1	About Linear Maps	11
2.2	Writing Linear Maps using Bases/Coordinates	13
2.2.1	Entries of the Representation Matrix	15
2.2.2	Representing Composition of Linear Maps by Matrices	15
3	Matrices	16
3.1	col-rank = row-rank	21
3.2	More on Matrices and Linear Equations	22
3.3	Elementary Row Operations Revisited	23
4	Constructing New Vector Spaces Out of Old Ones	24
4.0.1	Rings	25
4.1	Dual Space	25
4.2	The Dual Map	26
4.3	Reflexivity	28
4.4	Naturality	28
4.5	Direct Sums	29
4.5.1	Generalization to more Summands	30
4.6	Quotient Spaces	30
4.6.1	Universal Property of Quotient Spaces	32
4.6.2	Quotient Spaces & Complements	33
4.7	Relations to Dual Spaces	33
5	Determinants	33
5.1	Uniqueness of the Determinant	35
5.1.1	Important Thing About Permutations	36
5.1.2	Back to our $D(A)$	37
5.1.3	A Bit more on Permutations	38
5.2	More on Determinants	39
5.2.1	Determinants of Block Matrices	40
5.3	Cramer's Rule	41
5.4	Determinants of Endomorphisms	42
6	Eigenvalues & Eigenvectors	43

6.1	The Characteristic Polynomial	44
6.1.1	A Quick Digression about Polynomials	47
6.2	Geometric & Algebraic Multiplicities of Eigenvalues	48
6.3	Diagonalizability	49
6.4	Trigonalization	49
6.5	Minimal Polynomial & Cayley-Hamilton Theorem	50
7	Euclidean & Hermitian (unitary) Spaces	51
7.1	Norms & Angles	55
7.2	The Gram-Schmidt Orthogonalization Process	56
7.3	The Orthogonal Complement	57
7.4	Orthogonal & Unitary Matrices	60
7.5	QR-Decomposition	60
8	Dual Spaces & Inner Products	61
8.1	The Adjoint Map	62
8.2	Basic Properties of Adjoint Maps	63
8.3	Matrix Representation of Adjoint Maps	65
9	Spectral Theorems	65

1 Vector Spaces

1.1 Vector Spaces and Linear Subspaces

Definition 1.1: Vector Space

A **vector space** over a field K is a set V , endowed with two operations

$$\begin{aligned} + : V \times V &\rightarrow V, & (v_1, v_2) &\mapsto v_1 + v_2, \\ \cdot : K \times V &\rightarrow V, & (\alpha, v) &\mapsto \alpha \cdot v \end{aligned}$$

s.t. the following conditions (axioms) hold:

- (V1) $\forall v_1, v_2, v_3 \in V : v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$,
- (V2) $\exists$ an element $0 \in V : 0 + v = v \quad \forall v \in V$,
- (V3) $\forall v \in V \exists v' \in V : v + v' = 0$,
- (V4) $\forall v_1, v_2 \in V : v_1 + v_2 = v_2 + v_1$,
- (V5) $\forall a, b \in K \forall v \in V : a \cdot (b \cdot v) = (a \cdot b) \cdot v$,
- (V6) $\forall v \in V : 1 \cdot v = v$,
- (V7) $\forall a \in K \forall v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$,
- (V8) $\forall a, b \in K \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v$.

Lemma 1.2: Properties of Vector Spaces

Let V be a vector space over a field K . Then:

- (a) The 0 vector from (V2) is unique. To avoid confusion (e.g. with $0 \in K$) we will sometimes denote it by 0_V .
- (b) The element v' (corresponding to v) from (V3) is unique. We'll denote it by $-v$. We'll also write $w - v := w + (-v)$ for all $w \in V$,
- (c) $\forall v \in V : 0 \cdot v = 0$,
- (d) $\forall a \in K : a \cdot 0 = 0$,
- (e) $\forall v \in V : -1 \cdot v = -v$,
- (f) $\forall v \in V : -(-v) = v$,
- (g) If $a \cdot v = 0$ for some $a \in K, v \in V$, then either $a = 0$ or $v = 0$,
- (h) Associativity for $+$ holds also for some of n elements. $v_1 + \dots + v_n$ makes sense no matter how we put the brackets $v_1 + (v_2 + (\dots + (v_{n-1} + v_n)))$ or $(v_1 + v_2) + (\dots)$ etc.

Example: The Coordinate Space K^n

Let K be a field and $n \in \mathbb{N}$.

$$K^n = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

K^n is a vector space over K if we endow it with the following operations:

- Let $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in K^n$.

$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n),$$

- For $a \in K$

$$a \cdot v = a \cdot (a_1, \dots, a_n) := (a \cdot a_1, \dots, a \cdot a_n)$$

- $0 := (0, \dots, 0) \in K^n$

With these operations K^n becomes a vector space over K .

Definition 1.3: Linear Subspace

Let V be a vector space over K . A subset $W \subseteq V$ is called a **linear subspace** of V (or just subspace of V), if the following holds

(LSS1) $W \neq \emptyset$,

(LSS2) $\forall w_1, w_2 \in W : w_1 + w_2 \in W$,

(LSS3) $\forall a \in K, w \in W : a \cdot w \in W$.

Lemma 1.4: Linear Subspace is a Vector Space

Let V be a vector space over K and $W \subseteq V$ a subspace. Then W is a vector space on its own, when endowed with the operations from V .

Lemma 1.5: Criterion for a Linear Subspace

Let V be a vector space over K , and $W \subseteq V$ a subset. Then W is a linear subspace of V if and only if the following holds:

(1) $0_V \in W$,

(2) $\forall a_1, a_2 \in K, w_1, w_2 \in W : a_1 w_1 + a_2 w_2 \in W$.

Example: Spaces of functions

Fix a field K and a non-empty set S . Define

$$K^S = \{f : S \rightarrow K \mid f \text{ is a function}\}.$$

Define $+$ and $\cdot$ on K^S as follows. Let $f, g \in K^S$, $a \in K$. Define

$$\begin{aligned}(f + g) : S &\rightarrow K, & S \ni x &\mapsto f(x) + g(x), \\ (a \cdot f) : S &\rightarrow K, & S \ni x &\mapsto a \cdot f(x).\end{aligned}$$

Now Let $S = [0, 1]$ and take $K = \mathbb{R}$. Define $C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0, 1]\} \subseteq \mathbb{R}^S$. Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.

1.2 Span of Sets

Definition 1.6: Span

Let $S \subseteq V$ be a subset. We define

$$\text{Sp}(S) := \bigcap_{W \in \mathcal{N}} W,$$

where

$$\mathcal{N} := \{W \mid W \text{ is a subspace of } V \text{ and } S \subseteq W\},$$

i.e., the collection of all subspaces $W \subseteq V$ that contain S . $\text{Sp}(S)$ is called the span of S (in V).

Lemma 1.7: Properties of the Span

- (1) $\text{Sp}(S) \subseteq V$ is a linear subspace.
- (2) If $W \subseteq V$ is a linear subspace and $S \subseteq W$, then $\text{Sp}(S) \subseteq W$.

In other words, $\text{Sp}(S)$ is a linear subspace of V , and moreover among those subspaces that contain S , $\text{Sp}(S)$ is the smallest one.

Definition 1.8: Linear Combination

Let $n \in \mathbb{N}$, $a_1, \dots, a_n \in K$, $v_1, \dots, v_n \in V$. A **linear combination** of the vectors $v_1, \dots, v_n$ with coefficients $a_1, \dots, a_n$ is the vector

$$a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i \in V.$$

It is important to note that in this course consider only finitely many elements in the sums of linear combinations.

Lemma 1.9: Alternate Definition of Span

Let $\emptyset \neq S \subseteq V$. Then

$$\text{Sp}(S) = \{a_1 v_1 + \dots + a_n v_n \mid n \in \mathbb{N}, a_i \in K, v_i \in S \quad \forall 1 \leq i \leq n\}.$$

Remark 1.10. The set S can be finite or infinite. Regardless of that, each linear combination uses only finite numbers of elements from S .

Let $n \in \mathbb{N}$, $v_1, \dots, v_n \in V$. We'll write

$$\text{Sp}(v_1, \dots, v_n) := \text{Sp}(\{v_1, \dots, v_n\}).$$

Remark 1.11. *It obviously holds that*

$$\text{Sp}(v_1, \dots, v_n) = \left\{ \sum_{i=1}^n \alpha_i v_i \mid \alpha_i \in K \quad \forall 1 \leq i \leq n \right\}.$$

Remark 1.12. *It follows from the definition of span that*

$$\text{Sp}(\emptyset) = \{0_V\}.$$

Definition 1.13: Generating Set

Let V be a vect. sp. over K and $S \subseteq V$. We say that S **generates** (or **spans**) V if $\text{Sp}(S) = V$. S is called a generating set of V if $\text{Sp}(S) = W$, where $W \subseteq V$ is a subspace of V . We say that S **generates** (or **spans**) W .

Definition 1.14: Finite Dimensional

A vector space V is called **finite-dimensional** if $\exists$ a **finite** set $S \subseteq V$ such that $\text{Sp}(S) = V$. Otherwise V is called **infinite-dimensional**.

1.3 Bases of Vector Spaces

1.3.1 Linear Independence

Let V be a vector space over K .

Definition 1.15: Linear Independence

Let $v_1, \dots, v_n$ be a list of n vectors in V . We say that $v_1, \dots, v_n$ is **linearly independent** if the only linear combination of $v_1, \dots, v_n$ that gives $0 \in V$ is the trivial one. In other words,

$$\forall a_1, \dots, a_n \in K : a_1 v_1 + \dots + a_n v_n = 0 \implies a_1 = 0, \dots, a_n = 0.$$

By definition, we say that the empty list of vectors $\emptyset$ is linearly independent. We say that a list of vectors is linearly dependent if, the list is not linearly independent.

Remark 1.16. (1) 0 can always be written as a linear combination of any finite list of vectors $0v_1 + \dots + 0v_n = 0$, i.e., the trivial linear combination.

(2) If the list $v_1, \dots, v_n$ has two vectors that are the same, say $v_k = v_l$ for $1 \leq k < l \leq n$, then $v_1, \dots, v_n$ is **not** linearly independent. Indeed, $v_k + (-1)v_l = 0$.

(3) The order of the vectors in the list $v_1, \dots, v_n$ has no effect on linear independence. So, if there are no repetitions in $v_1, \dots, v_n$ we can talk about linear independence of the set $\{v_1, \dots, v_n\}$.

Definition 1.17: Linear Independence in the Context of Families

Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a family of vectors in V , indexed by a set I . We say that $\mathcal{F}$ is linearly independent if $\forall n \in \mathbb{N}$ and any sequence of distinct indices $i_1, \dots, i_n \in I$, the list of vectors $v_{i_1}, \dots, v_{i_n}$ is linearly independent. Of course, if in $\mathcal{F}$ there are any repetitions of vectors, i.e., if $\exists i, j, i \neq j$, s.t. $v_i = v_j$, then $\mathcal{F}$ cannot be linearly independent.

Lemma 1.18: Subsets of Linearly Independent Sets are Linearly Independent

If $\emptyset \neq S \subseteq V$ is linearly independent. Then every non-empty subset $S' \subseteq S$ is also linearly independent.

Definition 1.19: Linear Dependence

A family $\mathcal{F} = \{v_i\}_{i \in I}$ of vectors in V is called **linearly dependent** if it is not linearly independent.

1.3.2 Basis of a Vector Space**Definition 1.20: Basis of a Vector Space**

A subset $S \subseteq V$ of a vector space is called a basis of V if the following holds:

- (1) S is linearly independent.
- (2) S generates (or spans) V , i.e., $\text{Sp}(S) = V$.

Proposition 1.21: Equivalent Formulation of basis

A subset $S \subseteq V$ is a basis of V if and only if every $v \in V$ can be written in a unique way as a linear combination of vectors in S .

Proof. For simplicity we will assume that $|S| < \infty$. The case $|S| = \infty$ is similar.

$\Leftarrow$: We assume that every $v \in V$ can be written in a unique way as a linear combination of elements from S . Now, "can be written" means that $\text{Sp}(S) = V$. Next write $S = \{v_1, \dots, v_n\}$. Consider the vector $0 \in V$. Clearly

$$0 = 0 \cdot v_1 + \dots + 0 \cdot v_n.$$

Since 0 can be written in a unique way as a linear combination of $v_1, \dots, v_n$ it follows that

$$0 = a_1 v_1 + \dots + a_n v_n \quad \implies \quad a_1 = \dots = a_n = 0.$$

Therefore, S is linearly independent, which proves that S is a basis of V .

$\Rightarrow$: We assume that S is a basis for V . By assumption $\text{Sp}(S) = V$, so this means every $v \in V$ can be expressed as a linear combination of vectors from S . It remains to show that this unique.

Assume by contradiction that $\exists v \in V$ which has two different ways to be written as a linear combination of vectors from S , i.e.,

$$v = a_1 v_1 + \dots + a_n v_n = b_1 v_1 + \dots + b_n v_n$$

and $a_k \neq b_k$ for some $1 \leq k \leq n$. Then

$$0 = (a_1 - b_1)v_1 + \dots + \underbrace{(a_k - b_k)}_{\neq 0} + \dots + (a_n - b_n)v_n$$

and we deduce that 0 can be written as a non-trivial linear combination of elements from S . Therefore, S is linearly dependent, which is a contradiction to the fact that S is a basis of V . $\square$

1.3.3 Example: Bases

$V = M_{m \times n}(K)$, $\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis, where

$$E_{ij} = \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix},$$

where the 1 is at the position (i, j) in the matrix.

Lemma 1.22: A

Let $v_1, \dots, v_m \in V$ be a list of vectors which is linearly dependent. Then $\exists 1 \leq j \leq m$ s.t.:

- (1) $v_j \in \text{Sp}(v_1, \dots, v_{j-1})$.
- (2) $\text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m) = \text{Sp}(v_1, \dots, v_m)$. In other words, we can drop one of the vectors from the list without changing the span.

Lemma 1.23: B

Same assumptions as in Lemma 1.22, but we assume in addition that $v_1, \dots, v_k$ are linearly independent, where $1 \leq k \leq m$ is some index. Then $k < j$.

Lemma 1.24: C

Let $w_1, \dots, w_n \in V$ s.t. $\text{Sp}(w_1, \dots, w_n) = V$, and let $v \in V$. Then $v, w_1, \dots, w_n$ is linearly dependent.

Lemma 1.25: D

Let $v_1, \dots, v_n \in V$ be s.t. $\text{Sp}(v_1, \dots, v_n) = V$, and let $u_1, \dots, u_m \in V$ be linearly independent. Then $m \leq n$. Moreover, one can delete m vectors from the list $v_1, \dots, v_n$ s.t. the remaining list, say $v'_1, \dots, v'_{n-m}$ together with $u_1, \dots, u_m$ still span V , i.e., $\text{Sp}(u_1, \dots, u_m, v'_1, \dots, v'_{n-m}) = V$.

Lemma 1.26: E

Let $w_1, \dots, w_l \in V$ and assume that $w_j \notin \text{Sp}(w_1, \dots, w_{j-1}) \quad \forall 1 \leq j \leq l$. Then the list $w_1, \dots, w_l$ is linearly independent.

Lemma 1.27: F

Suppose that $\text{Sp}(v_1, \dots, v_n) = V$. Then $\exists$ a subset of $\{v_1, \dots, v_n\}$ which is a basis for V .

Theorem 1.28: Bases Have the Same Number of Elements

Let V be a finite dimensional vector space over K . Then V has a basis with finitely many elements. Moreover, every basis of V is finite and has the same number of elements.

Definition 1.29: Dimension of a Vector Space

Let V be a finite dimensional vector space over K . We define the dimension of V to be the number $n \in \mathbb{N}$ of elements in a basis (any basis !) of V . We write $\dim V := n$. In case V is not finite dimensional we write $\dim V = \infty$.

Remark 1.30. • $\dim V$ is well defined as it does not depend on a specific choice of basis for V .

- $\dim\{0\} = 0$, b.c. $\emptyset$ is a basis for the space $\{0\}$, and $|\emptyset| = 0$.

Summary

Let V be a finite dimensional vector space over K . Then:

- (1) Every finite list of vectors that spans V contains a sublist which is a basis for V .
- (2) Every linearly independent list can be extended to a basis of V .

Theorem 1.31: Equivalent Statements for Bases

Let V be a vector space of $\dim V = n \in \mathbb{N}$. Let $v_1, \dots, v_n \in V$. Then the following statements are equivalent:

- (1) $v_1, \dots, v_n$ are linearly independent.
- (2) $v_1, \dots, v_n$ generate V , i.e., $\text{Sp}(v_1, \dots, v_n) = V$.
- (3) $v_1, \dots, v_n$ is a basis for V .

Theorem 1.32: Consequences of Bases

Suppose that $n = \dim V < \infty$. Let $v_1, \dots, v_k \in V$. Then:

- (1) If $k < n$, the list $v_1, \dots, v_k$ **cannot** span V .
- (2) If $k > n$, the list $v_1, \dots, v_k$ **cannot** be linearly independent.

Proposition 1.33: Dimension of Subspaces

Let V be a finite dimensional vector space over K . Then every subspace $U \subseteq V$ is also finite dimensional. Moreover, $\dim U \leq \dim V$ with equality if and only if $U = V$.

1.4 Row and Column Spaces

Row and Column Spaces

Definition 1.34: Row and Column Space

Let $A \in M_{m \times n}(K)$ and $u_1, \dots, u_m \in K^n$ the rows of A and $v_1, \dots, v_n \in K^m$ the columns of A , i.e.,

$$A = \begin{pmatrix} - & u_1 & - \\ & \vdots & \\ - & u_m & - \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

We define two spaces

$$\text{RowS}(A) := \text{Sp}(u_1, \dots, u_m) \subseteq K^n$$

$$\text{ColS}(A) := \text{Sp}(v_1, \dots, v_n) \subseteq K^m.$$

Lemma 1.35: Row Equivalent Matrices

If $A, B \in M_{m \times n}(K)$ are row-equivalent matrices then $\text{RowS}(A) = \text{RowS}(B)$.

Remark 1.36. The statement of the Lemma does **not** hold for the column space.

1.4.1 Transpose of a Matrix

Definition 1.37: Transpose of a Matrix

Let $A = (a_{i,j}) \in M_{m \times n}(K)$. We define the transposed matrix of A as $A^T := (b_{i,j}) \in M_{n \times m}(K)$, where $b_{i,j} = a_{j,i}$.

Lemma 1.38: Properties of the Transpose

or every $A, B \in M_{m \times n}(K)$, $\lambda \in K$ we have

1. $(A + B)^T = A^T + B^T$.
2. $(\lambda A)^T = \lambda A^T$.
3. $(A^T)^T = A$.
4. $\text{RowS}(A^T) = \text{ColS}(A)$, $\text{ColS}(A^T) = \text{RowS}(A)$.

1.5 Sums of Vector Spaces

Definition 1.39: Sum of Vector Spaces

Let V be a vector space over K and $U, W \subseteq V$ linear subspaces. We define the sum $U + W$ of U, W as

$$U + W := \{u + w \mid u \in U, w \in W\}.$$

In a similar way we can define sums of more than two subspaces. If $U_1, \dots, U_r \subseteq V$ are subspaces, define

$$U_1 + \dots + U_r := \{u_1 + \dots + u_r \mid u_1 \in U_1, \dots, u_r \in U_r\}.$$

Proposition 1.40: Properties of the Sum of Vector Spaces

Let $U, W \subseteq V$ be linear subspaces. Then:

1. $U + W = \text{Sp}(U \cup W)$. In particular, $U + W \subseteq V$ is a subspace and $U, W \subseteq U + W$ are subspaces of $U + W$.
2. If U, W are finite dimensional then so is $U + W$. Moreover, one can find a basis for $U + W$ as follows:
 - (a) Choose a basis $p_1, \dots, p_k$ for $U \cap W$, where $k = \dim(U \cap W)$.
 - (b) Extend $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, u_1, \dots, u_{l-k}$ of U , where $l = \dim U$.
 - (c) Extend $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, w_1, \dots, w_{m-k}$ of W , where $m = \dim W$.

Then $p_1, \dots, p_k, u_1, \dots, u_{l-k}, w_1, \dots, w_{m-k}$ is a basis for $U + W$.

3. In particular, we have $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Corollary 1.41: Consequences of the Properties

Let $U, W \subseteq V$ be two finite dimensional linear subspaces of a vector space V . The following are equivalent:

1. $\dim(U + W) = \dim U + \dim W$.
2. $\dim(U \cap W) = 0$.
3. $U \cap W = \{0\}$.
4. Every $v \in U + W$ can be written in a **unique** way as $v = u + w$ with $u \in U, w \in W$.
5. If $u + w = 0$, where $u \in U$ and $w \in W$, then $u = w = 0$.

Definition 1.42: Linear Complement

Let V be a vector space over K and $U \subseteq V$ a linear subspace. A linear subspace $W \subseteq V$ is called a **linear complement** of U if $U + W = V$ and $U \cap W = \{0\}$. Note that U, W satisfy all the five equivalent statements from the corollary 1.41.

Proposition 1.43: Existence of a Complement

Let V be a finite dimensional vector space and $U \subseteq V$ a subspace. Then there exists a subspace $W \subseteq V$ which is a complement to U .

2 Linear Maps

Definition 2.1: Linear Map

Let V, W be vector spaces over K . A map $T : V \rightarrow W$ is called linear map if:

1. $\forall v_1, v_2 \in V : T(v_1 + v_2) = T(v_1) + T(v_2)$
2. $\forall \alpha \in K, \forall v \in V : T(\alpha \cdot v) = \alpha \cdot T(v)$

This means that $T : V \rightarrow W$ is a map that respects the structures of vector spaces on $(V, +, \cdot), (W, +, \cdot)$. We denote the set of all linear maps $T : V \rightarrow W$ by $\text{Hom}(V, W)$. Sometimes $\text{hom}_K(V, W)$.

Proposition 2.2: Properties of Linear Maps

Let V, W be vector spaces over a field K , and let $T : V \rightarrow W$ be a linear map. Then:

1. $\forall n \in \mathbb{N}, a_1, \dots, a_n \in K, v_1, \dots, v_n \in V$ we have

$$T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T(v_i).$$

2. $T(0_V) = 0_W$.

Theorem 2.3: A unique Linear Map

Let V be a finite dimensional vector space and $v_1, \dots, v_n \in V$ a basis for V . Let $w_1, \dots, w_n \in W$ be arbitrary vectors. Then there exists a unique linear map $T : V \rightarrow W$ s.t.

$$T(v_i) = w_i \quad \forall 1 \leq i \leq n.$$

Lemma 2.4: Matrix Representation of the Unique Linear Map

For every linear transformation $T : K^n \rightarrow K^m$ there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $T = T_A$.

2.1 Kernel & Image

Definition 2.5: Kernel and Image

Let V, W be vector spaces over K and $T : V \rightarrow W$ a linear map.

1. The set

$$\ker(T) := \{v \in V \mid Tv = 0\} \subseteq V$$

is a linear subspace of V . It is called the **kernel of T** .

2. The set

$$\text{Im}(T) := \{Tv \mid v \in V\} \subseteq W$$

is a linear subspace of W . It is called the **image of T** .

With the notation from set theory, we have

$$\ker(T) = T^{-1}(\{0\}), \quad \text{Im}(T) = T(V).$$

Proposition 2.6: Equivalences for Injective and Surjective

Let $T : V \rightarrow W$ be a linear map. Then:

1. T is injective $\iff \ker(T) = \{0\}$.
2. T is surjective $\iff \text{Im}(T) = W$.

2.1.1 About Linear Maps

Definition 2.7: Isomorphism

A linear map $T : V \rightarrow W$ is called an **isomorphism** if there exists a linear map $S : W \rightarrow V$ s.t. $S \circ T = \text{id}_V$, $T \circ S = \text{id}_W$. We say that V is isomorphic to W if there exists an isomorphism $T : V \rightarrow W$. We write $V \cong W$.

Lemma 2.8: Linearity of the Inverse Map

Let $T : V \rightarrow W$ be a linear map which is bijective. Then the inverse map $T^{-1} : W \rightarrow V$ is linear. In other words, a linear bijective map is an isomorphism.

Lemma 2.9: Composition of Linear Maps

Let $T : V \rightarrow W$ and $S : W \rightarrow U$ be linear maps. Then the composition $S \circ T : V \rightarrow U$ is also linear.

Definition 2.10: Endomorphism / Automorphism

A linear map $T : V \rightarrow V$ is sometimes called an **endomorphism** of V . We denote the set of all endomorphisms of V by $\text{End}(V)$. (So, $\text{End}(V) = \text{Hom}(V, V)$.)

A linear map $T : V \rightarrow V$ which is an isomorphism is also called an **automorphism** of V .

Lemma 2.11: Span of a Basis

Let $T : V \rightarrow W$ be a linear map. Let $v_1, \dots, v_n$ be a basis of V . Then $\text{Im}(T) = \text{Sp}(Tv_1, \dots, Tv_n)$

Theorem 2.12: Rank Theorem

Let V, W be vector spaces, where V is finite dimensional and let $T : V \rightarrow W$ be a linear map. Then

$$\dim \ker(T) + \dim \text{Im}(T) = \dim V.$$

Proof. Define $n := \dim V$. Let $u_1, \dots, u_k$ be a basis for $\ker(T)$. Note that $\ker(T)$ is finite dimensional, b.c. it is a subspace of V , which is finite dimensional. Extend $u_1, \dots, u_k$ to a basis $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ of V , which we can do by a previous result. By Lemma 2.11, we have that

$$\text{Im}(T) = \text{Sp}(\underbrace{T u_1, \dots, T u_k}_{=0}, T v_1, \dots, T v_{n-k}) = \text{Sp}(T v_1, \dots, T v_{n-k}).$$

We'll show now that $T v_1, \dots, T v_{n-k}$ is linearly independent. To see this, assume that

$$\sum_{i=1}^{n-k} a_i T v_i = 0,$$

for some $a_1, \dots, a_{n-k} \in K$. Since T is linear we can write this linear combination as

$$T\left(\sum_{i=1}^{n-k} a_i v_i\right) = 0 \quad \implies \quad \sum_{i=1}^{n-k} a_i v_i \in \ker(T).$$

Now, $u_1, \dots, u_k$ form a basis for $\ker(T)$. So, $\exists b_1, \dots, b_k \in K$ s.t.

$$\begin{aligned} \sum_{i=1}^{n-k} a_i v_i &= b_1 u_1 + \dots + b_k u_k \\ \implies b_1 u_1 + \dots + b_k u_k + (-a_1) v_1 + \dots + (-a_{n-k}) v_{n-k} &= 0. \end{aligned}$$

Since, $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ form a basis for V , we have that $b_1 = \dots = b_k = a_1 = \dots = a_{n-k} = 0$. This shows that $T v_1, \dots, T v_{n-k}$ are linearly independent. Therefore, $T v_1, \dots, T v_{n-k}$ form a basis for $\text{Im}(T)$. Thus,

$$\begin{aligned} \dim \text{Im}(T) &= n - k = \dim V - \dim \ker(T) \\ \implies \dim \ker(T) + \dim \text{Im}(T) &= \dim V. \end{aligned}$$

□

Corollary 2.13: Consequences of the Rank Theorem

Let $T : V \rightarrow W$ be a linear map, where V, W are finite dimensional vector spaces. Then the following holds:

1. If $\dim W < \dim V$ then T is **not** injective.
2. If $\dim W > \dim V$ then T is **not** surjective.
3. If $\dim W = \dim V$ then T is bijective $\iff T$ is surjective $\iff T$ is injective.

Corollary 2.14: Equivalence for an Isomorphism

Two finite dimensional vector spaces V, W are isomorphic if and only if $\dim V = \dim W$.

Theorem 2.15: Image of a Set

Let $T : V \rightarrow W$ be an isomorphism between two vector spaces V, W . Let S be a family of vector in V . Write $T(S) := \{Tv\}_{v \in S}$. Then:

1. S is linearly independent if and only if $T(S)$ is linearly independent.
2. S spans V if and only if $T(S)$ spans W .
3. S is a basis of V if and only if $T(S)$ is a basis of W .

What is the significance of isomorphisms? If two vector spaces V, W are isomorphic, $V \cong W$ (i.e., $\exists T : V \rightarrow W$ bijection), then V and W have the same linear algebraic properties (like dimension). We can think of V and W as the same space represented in two different ways.

However, identifying V with W usually requires a choice of an isomorphism $T : V \rightarrow W$. Usually there is no preferred (canonical) choice. It is therefore better, in general, not to identify different vector spaces that are isomorphic.

Definition 2.16: Rank

Let $T : V \rightarrow W$ be a linear map. We define the **rank** of T by

$$\text{rank}(T) := \dim \text{Im}(T)$$

Lemma 2.17: Rank of the Composition

Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps, where U, V, W are finite dimensional vector spaces. Then

1. $\text{rank}(S \circ T) \leq \min\{\text{rank}(S), \text{rank}(T)\}$.
2. If S is injective, then $\text{rank}(S \circ T) = \text{rank}(T)$.
3. If T is surjective, then $\text{rank}(S \circ T) = \text{rank}(S)$.

2.2 Writing Linear Maps using Bases/Coordinates

In this Section we'll assume all vector spaces to be finite dimensional unless otherwise stated. We'll write bases of a vector space V as an ordered list $\mathcal{B} = (v_1, \dots, v_n)$.

Definition 2.18: Coordinate Map

Let V be a vector space over K and $\mathcal{B} = (v_1, \dots, v_n)$ a basis of V . Let $v \in V$. We define the coordinate-vector of v w.r.t. the basis $\mathcal{B}$ as

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \dots, a_n \in K$ are the unique scalars in K s.t. $v = a_1 v_1 + \dots + a_n v_n$. Note that $[v]_{\mathcal{B}}$ is uniquely defined since every $v \in V$ has a unique representation as a linear combination of $v_1, \dots, v_n$.

Define a map $\Phi_{\mathcal{B}} : V \rightarrow K^n$ as

$$\Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \in K^n.$$

Proposition 2.19: $\Phi_{\mathcal{B}}$ is an Isomorphism

$\Phi_{\mathcal{B}}$ is an isomorphism

Remark 2.20. The inverse map to $\Phi_{\mathcal{B}}$ is given by

$$\Phi_{\mathcal{B}}^{-1} : K^n \rightarrow V, \quad \Phi_{\mathcal{B}}^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} := a_1 v_1 + \dots + a_n v_n.$$

Corollary 2.21: Isomorphism Coordinate Space

Let V be a vector space over K with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Then $V \cong K^n$.

Proof. Choose a basis $\mathcal{B}$ for V . Then $\Phi_{\mathcal{B}} : V \rightarrow K^n$ is an isomorphism. □

Remark 2.22. 1. The corollary does not hold for $\dim V = \infty$, at least not as stated.

2. We've seen that $V \cong K^n$, if V is finite dimensional, but in general there does **not** exist a canonical isomorphism $V \xrightarrow{\cong} K^n$. For example $\Phi_{\mathcal{B}}$ depends on the choice of basis $\mathcal{B}$.

Now we arrive at the main question. Let V, W be vector spaces over K , $n := \dim V, m := \dim W$, and let $T : V \rightarrow W$ be a linear map. Fix a basis $\mathcal{B}$ for V and a basis $\mathcal{C}$ for W . Consider the diagram

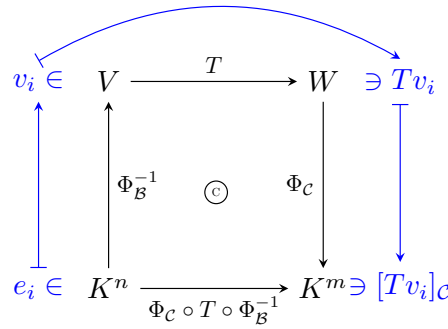
$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \Phi_{\mathcal{B}}^{-1} \uparrow & \text{\textcircled{C}} & \downarrow \Phi_{\mathcal{C}} \\ K^n & \xrightarrow{\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}} & K^m \end{array}$$

The 'C' means that the diagram is commutative.

Since Φ_B^{-1}, T, Φ_C are all linear maps, the map $\Phi_C \circ T \circ \Phi_B^{-1} : K^n \rightarrow K^m$ is also linear. By Lemma 2.4, any linear map $K^n \rightarrow K^m$ can be written using a matrix $A \in M_{m \times n}(K)$. In other words, there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $\Phi_C \circ T \circ \Phi_B^{-1} = T_A$. The matrix A is called the **representation matrix of T w.r.t. the bases $\mathcal{B}$ and $\mathcal{C}$** . We denote this matrix by $[T]_{\mathcal{C}}^{\mathcal{B}} \in M_{m \times n}(K)$.

2.2.1 Entries of the Representation Matrix

Let $T : V \rightarrow W$ be a linear map and $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V , and $\mathcal{C} = (w_1, \dots, w_m)$ a basis for W .



$T_A(e_i) = \Phi_C \circ T \circ \Phi_B^{-1}(e_i) = \Phi_C \circ T v_i = [T v_i]_{\mathcal{C}}$. But these are precisely the columns of A . In other words

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T v_1]_{\mathcal{C}} & \dots & [T v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix} \in M_{m \times n}.$$

Another way to describe this matrix is: Let $A = (a_{i,j})$, where $a_{i,j}$ are defined uniquely by

$$T v_j = \sum_{i=1}^m a_{i,j} w_i.$$

Proposition 2.23: Representation matrix

Let $T : V \rightarrow W$ be a linear map between two finite dimensional vector spaces V, W . Let $\mathcal{B}, \mathcal{C}$ be bases for V, W respectively. Then

$$\forall v \in V : [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T v]_{\mathcal{C}}.$$

2.2.2 Representing Composition of Linear Maps by Matrices

Let $T : V \rightarrow W, S : W \rightarrow U$ be linear maps. Let $\mathcal{A} = (v_1, \dots, v_n)$ be a basis for V , $\mathcal{B} = (w_1, \dots, w_m)$ a basis for W and $\mathcal{C} = (u_1, \dots, u_p)$ a basis for U .

Write $[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij})$, $[S]_{\mathcal{C}}^{\mathcal{B}} = (b_{ij})$, $[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = (c_{ij})$. By definition we have that

$$T(v_j) = \sum_{k=1}^m a_{kj} w_k \quad \forall 1 \leq j \leq n \quad (2.1)$$

$$T(w_j) = \sum_{i=1}^p b_{ij} u_i \quad \forall 1 \leq j \leq m \quad (2.2)$$

$$S \circ T(v_j) = \sum_{i=1}^p c_{ij} u_i \quad \forall 1 \leq j \leq n \quad (2.3)$$

But we can calculate $S \circ T(v_j)$ by using Equations (2.1), (2.2). Then we have

$$\begin{aligned} S \circ T(v_j) &= S\left(\sum_{k=1}^m a_{kj} w_k\right) = \sum_{k=1}^m a_{kj} S(w_k) = \sum_{k=1}^m a_{kj} \left(\sum_{i=1}^p b_{ik} u_i\right) \\ &= \sum_{k=1}^m \sum_{i=1}^p a_{kj} b_{ik} u_i = \sum_{i=1}^p \sum_{k=1}^m a_{kj} b_{ik} u_i = \sum_{i=1}^p \left(\sum_{k=1}^m b_{ik} a_{kj}\right) u_i \\ \implies c_{ij} &= \sum_{k=1}^m b_{ik} a_{kj}. \end{aligned}$$

3 Matrices

Definition 3.1: Matrix Multiplication

Let $A \in M_{m \times n}(K)$, $B \in M_{n \times p}(K)$. We define a new matrix $A \cdot B \in M_{m \times p}(K)$ which is called the product of A and B . Write $A = (a_{ij})$, $B = (b_{ij})$. The matrix $A \cdot B$ has entries (c_{ij}) , where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Important: $A \cdot B$ is defined only when the number of columns of A equals the number of rows of B .

Conclusion: Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps between finite dimensional vector spaces. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be bases for V, W, U respectively. Then

$$[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = [S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}}.$$

Definition 3.2: Kronecker-Delta/Identity Matrix

Let $\mathcal{I}$ be an index set. Define for all $i, j \in \mathcal{I}$

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

This is called the **kronecker-delta**.

We define the **identity matrix** $I_n \in M_{n \times n}(K)$ as

$$I_n := \delta_{ij}, \quad I_n = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

Remark 3.3. Let V be a finite dimensional vector space and $n := \dim V$. Let $\text{id} : V \rightarrow V$ be the identity map. Let $\mathcal{B}$ be any basis for V . Then $[\text{id}]_{\mathcal{B}}^{\mathcal{B}} = I_n$. In other words, the matrix representation of the identity map w.r.t $\mathcal{B}$ is the identity matrix I_n .

Proposition 3.4: Associativity and Distributivity of Matrix Multiplication

$$1. \forall A \in M_{m \times n}(K), B \in M_{n \times p}(K), C \in M_{p \times q}(K) : (AB)C = A(BC) \in M_{m \times q}.$$

$$2. \forall A, B \in M_{m \times n}(K), C \in M_{n \times p}(K), C' \in M_{q \times m}(K) \text{ we have}$$

$$(A + B)C = AC + BC \in M_{m \times p}(K),$$

$$C'(A + B) = C'A + C'B \in M_{q \times n}(K).$$

$$3. \forall A \in M_{m \times n}(K) : I_m A = A I_n = A.$$

$$4. \forall \alpha \in K, A \in M_{m \times n}(K), B \in M_{n \times p}(K) : \alpha(AB) = (\alpha A)B = A(\alpha B).$$

Remark 3.5. Matrix multiplication is in general **not** commutative, i.e., $\exists A, B$ s.t. $AB \neq BA$.

In $M_{n \times n}(K)$ we can add/subtract matrices and also multiply matrices. There is also a unit I_n . These operations satisfy properties like associativity, distributivity, etc. From this we get that $M_{n \times n}(K)$ is a (non-commutative) **ring** with a unity.

Definition 3.6: Invertibility

An $n \times n$ matrix $A \in M_{n \times n}(K)$ is **invertible** if $\exists B \in M_{n \times n}(K)$ s.t. $AB = BA = I_n$.

Remark 3.7. If A is invertible, then there exists only one matrix B s.t. $AB = BA = I_n$.

Proof. Assume $AB = BA = I_n$ and $AC = CA = I_n$. Then

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

□

Definition 3.8: The Inverse Matrix

If A is invertible, the unique matrix B s.t. $AB = BA = I_n$ is called **the inverse of A** and is denoted by A^{-1} . So, we have $AA^{-1} = A^{-1}A = I_n$.

We denote the set of all invertible $n \times n$ matrices by

$$GL_n(K) = GL(n, K) := \{A \in M_{n \times n}(K) \mid A \text{ is invertible}\}.$$

This set is called the general linear group.

Proposition 3.9: Structure of the General Linear Group

$GL_n(K)$, endowed with matrix multiplication $(A, B) \mapsto A \cdot B$ is a group. The unity is I_n . The inverse of $A \in GL_n(K)$ is A^{-1} . Moreover,

$$\forall A, B \in GL_n(K) : (A^{-1})^{-1} = A, \quad (AB)^{-1} = B^{-1}A^{-1}.$$

Corollary 3.10: Unique Solution for Linear System of Equations

Let $A \in GL_n(K)$. Then $\forall b \in K^n$ the linear system of equations $Ax = b$ has a unique solution, namely $x = A^{-1}b$.

Proposition 3.11: Invertibility and Isomorphisms

Let $A \in M_{n \times n}(K)$. Then $A \in GL_n(K)$ if and only if $T_A : K^n \rightarrow K^n$ is an isomorphism. Moreover, $(T_A)^{-1} = T_{A^{-1}}$.

Definition 3.12: Upper/Lower Triangular and Diagonal Matrices

A matrix $A \in M_{n \times n}(K)$, $A = (a_{ij})$, is called

- **Upper triangular** if $a_{ij} = 0 \quad \forall i > j$, i.e.,

$$\begin{pmatrix} * & \dots & \dots & * \\ 0 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}.$$

- **Lower Triangular** if $a_{ij} = 0 \quad \forall i < j$, i.e.,

$$\begin{pmatrix} * & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & 0 \\ * & \dots & \dots & * \end{pmatrix}.$$

- **Diagonal** if $a_{ij} = 0 \quad \forall i \neq j$, i.e.,

$$\begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

Lemma 3.13: Multiplication of Triangular/Diagonal Matrices

Let A, B be two diagonal (respectively both upper triangular, respectively both lower triangular). Then $A \cdot B$ is of the same type.

Corollary 3.14: Change of Bases for a Linear Map

Let V, W be finite dimensional vector spaces over K . Let $n := \dim V, m := \dim W$. Let $\mathcal{B}, \mathcal{B}'$ be two bases for V , and $\mathcal{C}, \mathcal{C}'$ be two bases for W . Then

$$[id_V]_{\mathcal{B}}^{\mathcal{B}'} \in GL_n(K), [id_W]_{\mathcal{C}'}^{\mathcal{C}} \in GL_m(K), \text{ and} \\ [id_V]_{\mathcal{B}}^{\mathcal{B}'} = ([id_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}, [id_W]_{\mathcal{C}}^{\mathcal{C}'} = ([id_W]_{\mathcal{C}'}^{\mathcal{C}})^{-1}.$$

Let $T : V \rightarrow W$ be a linear map. Then

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [id_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}'}.$$

Definition 3.15: Transition Matrix/Change of Bases Matrix

Let V be a finite dimensional vector space over K , $n := \dim V$. Let $\mathcal{B}, \mathcal{B}'$ be two bases of V . The matrix $[id_V]_{\mathcal{B}'}^{\mathcal{B}} \in GL_n(K)$ is called the **change of bases matrix** between $\mathcal{B}$ and $\mathcal{B}'$, or the **transition matrix** from basis $\mathcal{B}$ to basis $\mathcal{B}'$.

If $\mathcal{B} = (v_1, \dots, v_n), \mathcal{B}' = (v'_1, \dots, v'_n)$ then

$$[id_V]_{\mathcal{B}'}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{B}'} & \dots & [v_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}.$$

Proposition 3.16: Inverse of a Matrix with Bases

Let $T : V \rightarrow W$ be an isomorphism between two finite dimensional vector spaces. Let $\mathcal{B}, \mathcal{C}$ be bases for V, W respectively. Then $[T]_{\mathcal{C}}^{\mathcal{B}}$ is an invertible matrix and

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^{-1}.$$

Corollary 3.17: Change of Basis for an Endomorphism

Let V be a finite dimensional vector space and $T : V \rightarrow V$ a linear map (an endomorphism). Let $\mathcal{B}, \mathcal{B}'$ be two bases for V . Then

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'} = ([id_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'}.$$

Definition 3.18: Equivalence and Similarity of Matrices

1. Let $A, B \in M_{m \times n}(K)$. We say that A and B are **equivalent** if $\exists P \in GL_m(K), Q \in GL_n(K)$ s.t. $B = PAQ$.
2. Let $A, B \in M_{n \times n}(K)$. We say that A and B are **similar** if $\exists P \in GL_n(K)$ s.t. $B = P^{-1}AP$.

Remark 3.19. 1. Equivalence between $m \times n$ matrices gives an equivalence relation on $M_{m \times n}(K)$.

2. $A, B \in M_{m \times n}(K)$ are equivalent if and only if there exists two vector spaces V, W with $n := \dim V$, $m := \dim W$ and two bases $\mathcal{B}, \mathcal{B}'$ of V and two bases $\mathcal{C}, \mathcal{C}'$ for W s.t. $A = [T]_{\mathcal{C}}^{\mathcal{B}}$, $B = [T]_{\mathcal{C}'}^{\mathcal{B}'}$. In other words, A is equivalent to B if and only if A and B are matrix representations of the same linear map w.r.t. different choices of bases for V and W .

3. Similarity between $n \times n$ matrices gives an equivalence relation on $M_{n \times n}(K)$.

4. $M, N \in M_{n \times n}(K)$ are similar if and only if there exists an n -dimensional vector space V and $T : V \rightarrow V$ and two bases $\mathcal{B}, \mathcal{B}'$ of V s.t. $M = [T]_{\mathcal{B}}^{\mathcal{B}}$, $N = [T]_{\mathcal{B}'}^{\mathcal{B}'}$.

Proposition 3.20: Existence of 'Nice' Bases

Let V, W be finite dimensional vector spaces over K , of dimension $n := \dim V$, $m := \dim W$. Let $T : V \rightarrow W$ be a linear map with $\text{rank}(T) = r$. (Recall that $\text{rank}(T) := \dim \text{Im}(T)$.) Then there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V and a basis $\mathcal{C} = (w_1, \dots, w_m)$ of W s.t.

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix}.$$

Corollary 3.21

Let $A \in M_{m \times n}(K)$. Then $\exists P \in GL_m(K), Q \in GL_n(K)$ s.t.

$$PAQ = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix},$$

where $r = \text{col-rank}(A)$. In particular, $\text{col-rank}(A) \leq \min\{m, n\}$.

Corollary 3.22: Condition for Equivalence of Matrices

Let $A, B \in M_{m \times n}(K)$. Then $A \sim B$ if and only if $\text{col-rank}(A) = \text{col-rank}(B)$.

Moreover, $\forall 0 < r \leq \min\{m, n\}$ there is precisely one equivalence class of matrices A with $\text{col-rank} = r$.

3.1 col-rank = row-rank**Theorem 3.23: col-rank = row-rank**

Let $A \in M_{m \times n}(K)$. Then $\text{col-rank}(A) = \text{row-rank}(A)$.

Remark 3.24. 1. Because of this Theorem we'll denote from now on $\text{rank}(A) := \text{col-rank}(A) = \text{row-rank}(A)$.

2. Suppose A' is a row-reduced echelon matrix which is row-equivalent to A . Then we have that $\text{rank}(A) = \# \text{ pivots in } A'$. Indeed, $\text{rank}(A) = \text{row-rank}(A) = \text{row-rank}(A') = \# \text{ pivots in } A'$.
3. Recall that for $T : V \rightarrow W$ we defined $\text{rank}(T) := \dim \text{Im}(T)$. Also, if $A \in M_{m \times n}(K)$, then $\text{rank}(T_A) = \dim \text{Im}(T) = \text{col-rank}(A)$.

We now present some Lemmas for the proof of the Theorem.

Lemma 3.25: Isomorphisms preserve the Rank

Let $T : V \rightarrow W$ be a linear map, and $S : W \xrightarrow{\cong} U$, $L : P \xrightarrow{\cong} V$ be isomorphisms. Then

1. $\text{rank}(S \circ T) = \text{rank}(T)$.
2. $\text{rank}(T \circ S) = \text{rank}(T)$.

Lemma 3.26

Let $A \in M_{m \times n}(K)$. Then A is invertible if and only if $T_A K^n \rightarrow K^n$ is an isomorphism.

Lemma 3.27: Equivalent Matrices Have the Same Rank

Let $A, B \in M_{m \times n}(K)$ s.t. $B = PAQ$, where $P \in GL_m(K)$, $Q \in GL_n(K)$. Then

1. $\text{col-rank}(A) = \text{col-rank}(B)$
2. $\text{row-rank}(A) = \text{row-rank}(B)$.

Proof. Proof of Theorem 3.23: By Corollary 3.21, $\exists P \in GL_m(K), Q \in GL_n(K)$ s.t.

$$PAQ = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix}.$$

Denote this matrix by B . For B we have $\text{col-rank}(B) = \text{row-rank}(B) = r$. By Lemma 3.27, we have that $\text{col-rank}(A) = \text{col-rank}(B) = r$, and $\text{row-rank}(A) = \text{row-rank}(B) = r$. Therefore, $\text{col-rank}(A) = \text{row-rank}(A)$. $\square$

Remark 3.28. Although $\text{row-rank}(A) = \text{col-rank}(A)$, the spaces $\text{ColS}(A)$ and $\text{RowS}(A)$ are **different**. First of all if $A \in M_{m \times n}$, then $\text{ColS}(A) \subseteq K^m$ and $\text{RowS}(A) \subseteq K^n$. But even when $n = m$, these two spaces are **different**. They have the same dimension, but are different.

3.2 More on Matrices and Linear Equations

Let $A \in M_{m \times n}(K)$, and consider the system of linear equations $Ax = 0$. Denote by

$$\text{Sol}(A, 0) := \{x \in K^n \mid Ax = 0\} = \ker(T_A),$$

the **space** of solutions $Ax = 0$. Note that $\dim \text{Sol}(A, 0) = n - \text{rank}(A)$ (b.c. $\dim \ker(T_A) = n - \dim \text{Im}(T_A)$).

Let A' be a row-reduced echelon matrix which is row-equivalent to A (e.g. A' is obtained from A by Gauss-elimination). # pivots of $A' = \text{rank}(A)$, $n - \text{rank}(A) = \#$ free variables. Also, we have that $\text{Sol}(A, 0) = \text{Sol}(A', 0)$.

$$\begin{pmatrix} 0 & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Denote by $x_{i_1}, \dots, x_{i_l}$ the free variables, where $1 \leq i_1 < \dots < i_l \leq n$, $l = n - \text{rank}(A)$ and by $x_{j_1}, \dots, x_{j_k}$, where $1 \leq j_1 < \dots < j_k \leq n$, $r = \text{rank}(A)$ the pivots. For each choice of values for the free variables $x_{i_1}, \dots, x_{i_l} \in K$, the pivot variables $x_{j_1}, \dots, x_{j_k}$ are uniquely determined. So, we obtain a map

$$\Phi : K^l \rightarrow \text{Sol}(A, 0)$$

$$\Phi(\lambda_1, \dots, \lambda_l) = \text{the unique solution of } Ax = 0,$$

with $x_{i_1} = \lambda_1, \dots, x_{i_l} = \lambda_l$.

Proposition 3.29: Φ is an Isomorphism

The map Φ is linear. Moreover, it is an isomorphism.

Corollary 3.30

Let $b \in K^m$, $A \in M_{m \times n}(K)$. Then

1. $\text{Sol}(A, b) := \{x \in K^n \mid Ax = b\} \neq \emptyset$ iff $b \in \text{ColS}(A)$.
2. If $\text{Sol}(A, b) \neq \emptyset$ and $y \in \text{Sol}(A, b)$ then

$$\text{Sol}(A, b) = y + \text{Sol}(A, 0) := \{y + x \mid x \in \text{Sol}(A, 0)\}.$$

Proposition 3.31

Let $A \in M_{m \times n}(K)$, $b \in K^m$. The following statements are equivalent:

1. $\text{rank}(A|b) = \text{rank}(A)$.
2. The system of equations $Ax = b$ has a solution.

Proposition 3.32

1. $\text{rank}(A|b) = \text{rank}(A) = n$.
2. The system of equations $Ax = b$ has a unique solution.

3.3 Elementary Row Operations Revisited

Let $A \in M_{n \times p}(K)$. We learned about elementary row operations which are used as part of the Gauss elimination. All of these operations can be expressed by matrix multiplications.

Denote by E_{ij} the matrix in Example 1.3.3.

Type I: Let $1 \leq i \leq n$, $1 \leq j \leq n$, $i \neq j$. Let $\alpha \in K$. Define $Q_{ij}(\alpha)$ to be the matrix obtained from I_n after applying $R_i + \alpha R_j \rightarrow R_i$. We have $Q_{ij}(\alpha) = I_n + \alpha E_{ij}$.

Type II: Let $1 \leq i \leq n$, $1 \leq j \leq n$. Define P_{ij} to be the matrix obtained from I_n by permuting (exchanging) row $\# i$ and row $\# j$, i.e. applying $R_i \leftrightarrow R_j$. We have $P_{ij} = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$.

Type III: Let $1 \leq i \leq n$ and $0 \neq \alpha \in K$. Denote by $S_i(\alpha)$ the matrix obtained from I_n by applying $\alpha R_i \rightarrow R_i$.

Lemma 3.33: Elementary Row Operations as Matrix Multiplication

Let $A \in M_{n \times p}(K)$.

1. Multiplying A from the left with $Q_{ij}(\alpha)$ results in the row operation $R_i + \alpha R_j \rightarrow R_i$: $A \xrightarrow{R_i + \alpha R_j \rightarrow R_i} Q_{ij}(\alpha)A$.
2. Multiplying A from the left with P_{ij} results in the row operation $R_i \leftrightarrow R_j$: $A \xrightarrow{R_i \leftrightarrow R_j} P_{ij}A$.
3. Multiplying A from the left with $S_i(\alpha)$ results in the row operation $\alpha R_i \rightarrow R_i$: $A \xrightarrow{\alpha R_i \rightarrow R_i} S_i(\alpha)A$.

The same holds for elementary column operations, which is done by right hand side multiplications of the matrices.

Lemma 3.34: Elementary Matrices are Invertible

Every elementary matrix is invertible, and the inverse of each of them is an elementary matrix of the same type, i.e.

$$Q_{ij}(\alpha)^{-1} = Q_{ij}(-\alpha), \quad P_{ij}^{-1} = P_{ji}, \quad S_{ij}(\alpha)^{-1} = S_{ij}(\alpha^{-1}).$$

Theorem 3.35

$\forall A \in GL_n(K)$, $\exists k \geq 1$ and elementary matrices $T_1, \dots, T_k$ s.t. $T_k \cdot \dots \cdot T_1 \cdot A = I_n$. Furthermore, we have that

$$A = T_1^{-1} \cdot \dots \cdot T_k^{-1}, \quad A^{-1} = T_k \cdot \dots \cdot T_1.$$

In particular, A can be written as a finite product of elementary matrices.

Theorem 3.36: Equivalent condition for Invertibility

Let $A \in M_{n \times n}(K)$. Place A together with I_n in the following way $(A|I_n)$. Then A is invertible if and only if $(A|I_n)$ can be brought via a sequence of elementary row-operations to the form $(I_n|B)$ for some matrix $B \in M_{n \times n}(K)$. In this case we have $A^{-1} = B$.

4 Constructing New Vector Spaces Out of Old Ones

Proposition 4.1: The Vector Space of Linear Functions

Let V, W be two vector spaces over K . Then $\text{Hom}(V, W)$ has the structure of a vector space over K , if we endow it with the following operations:

- $\forall T_1, T_2 \in \text{Hom}(V, W), (T_1 + T_2)(v) := T_1(v) + T_2(v) \quad \forall v \in V.$
- $\forall T \in \text{Hom}(V, W), \alpha \in K, (\alpha T)(v) := \alpha T(v) \quad \forall v \in V.$

The neutral element of $\text{Hom}(V, W)$ is the map $0 : V \rightarrow W, \quad v \mapsto 0.$

Theorem 4.2: The map $\Psi_C^{\mathcal{B}}$

Let V, W be finite dimensional vector spaces over K , $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V , and $\mathcal{C} = (w_1, \dots, w_m)$ a basis for W . Then the map

$$\Psi_C^{\mathcal{B}} : \text{Hom}(V, W) \rightarrow M_{m \times n}(K)$$

defined by

$$T \mapsto [T]_{\mathcal{C}}^{\mathcal{B}} := \begin{pmatrix} | & & | \\ [Tv_1]_{\mathcal{C}} & \dots & [Tv_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}$$

is a linear map and moreover an isomorphism.

Corollary 4.3: Dimension of $\text{Hom}(V, W)$

If V, W are finite dimensional vector spaces then so is $\text{Hom}(V, W)$ and $\dim \text{Hom}(V, W) = \dim V \cdot \dim W$.

4.0.1 Rings

Recall that Rings have the same axioms as Fields except that we do **not** require that every non-zero element has an inverse.

Definition 4.4: Homomorphism of Rings

Let R, S be rings. A **homomorphism of rings** is a map $\varphi : R \rightarrow S$ s.t.:

- $\forall r_1, r_2 \in R : \varphi(r_1 + r_2) = \varphi(r_1) + \varphi(r_2)$.
- $\forall r_1, r_2 \in R : \varphi(r_1 \cdot r_2) = \varphi(r_1) \cdot \varphi(r_2)$
- $\varphi(1) = 1$.
- $\varphi(0) = 0$.

Corollary 4.5: Ψ for an Endomorphism is an isomorphism of Rings

Let V be a vector space over K . Then $\text{End}(V) := \text{Hom}(V, V)$ is a ring with a unity. This ring is, in general, not commutative. The multiplication (product) in $\text{End}(V)$ is given by composition, i.e.,

$$T_1 \cdot T_2 := T_1 \circ T_2 \quad \forall T_1, T_2 \in \text{Hom}(V, V).$$

The unity is id_V . Moreover, if V is finite dimensional and $\dim V = n$, and $\mathcal{B} = (v_1, \dots, v_n)$ is a basis for V , then $\Psi_{\mathcal{B}}^{\mathcal{B}} : \text{Hom}(V, V) \rightarrow M_{n \times n}(K)$ is an isomorphism of rings

4.1 Dual Space

An important special case of $\text{Hom}(V, W)$ is when $W = K$.

Definition 4.6: Dual Space

Let V be a vector space over K . The **dual space** of V , is defined to be the vector space $V^* := \text{Hom}(V, K)$. The elements of V^* are linear maps $\ell : V \rightarrow K$, and are sometimes called **functionals** (on V).

Corollary 4.7: Dimension of the Dual Space

Let V be a finite dimensional vector space over K . Then $\dim V^* = \dim V$.

Let $V = K_{\text{col}}^n$. Then we can identify V^* with K_{col}^n , as follows: If $\ell \in K_{\text{row}}^n$, then ℓ defines a functional $f_{\ell} : K_{\text{col}}^n \rightarrow K$ by $f_{\ell}(v) := \ell \cdot v \in K$. The map $D : K_{\text{row}}^n \rightarrow (K_{\text{col}}^n)^*$ defined by $D(\ell) := f_{\ell}$ is a linear map and is an isomorphism.

Let V be a finite dimensional vector space over K and $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V . $\forall 1 \leq i \leq n$,

define a functional $v_i^* \in V^*$, i.e., $v_i^* : V \rightarrow K$ linear map, as follows:

$$v_i^*(v_j) := \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Since $v_1, \dots, v_n$ form a basis for V , our definition determines uniquely a well-defined functional $v_i^* : V \rightarrow K$.

Lemma 4.8: The Dual Basis

The elements $v_1^, \dots, v_n^* \in V^*$ form a basis for V^* . This basis is called the **dual basis** to $\mathcal{B}$. We denote this basis by $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$.*

Moreover,

$$\forall \ell \in V^* : \ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*.$$

Proof. Let $\ell \in V^*$. We claim that

$$\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*. \quad (4.1)$$

Or in other words, we claim that

$$\forall v \in V : \ell(v) = \sum_{i=1}^n \ell(v_i) \cdot v_i^*(v). \quad (4.2)$$

Indeed, both sides of Equation (4.1) are linear maps $V \rightarrow K$, so it's enough to check that Equation (4.2) holds for $v = v_1, \dots, v = v_n$, b.c. $v_1, \dots, v_n$ form a basis for V . And indeed, for $v = v_k$ we have

$$\sum_{i=1}^n \ell(v_i) \cdot v_i^*(v_k) = \sum_{i=1}^n \ell(v_i) \cdot \delta_{ik} = \ell(v_k).$$

The claim just proven shows that $v_1^*, \dots, v_n^*$ span V^* . Since $\dim V^* = \dim V = n$, it follows that $v_1^*, \dots, v_n^*$ form a basis for V^* . $\square$

Question: Let V be a finite dimensional vector space over K . Let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V . Therefore, $\mathcal{B}^*$ and $\mathcal{C}^*$ are two bases for V^* . What is the relation between $[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$ and $[id_V]_{\mathcal{B}}^{\mathcal{C}}$.

Proposition 4.9: Relation between $[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$ and $[id_V]_{\mathcal{B}}^{\mathcal{C}}$

$$[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([id_V]_{\mathcal{C}}^{\mathcal{B}})^T = \left(([id_V]_{\mathcal{B}}^{\mathcal{C}})^T \right)^{-1}.$$

4.2 The Dual Map

Definition 4.10: The Dual Map

Let V, W be vector spaces over K . Let $T : V \rightarrow W$ be a linear map. Define a new map

$$T^* : W^* \rightarrow V^*, \quad T^*(\ell) = \ell \circ T \quad \forall \ell \in W^*.$$

Then T^* is a linear map. It is called **the dual map** to T .

Proof. Let $\ell \in W^*$, i.e., $\ell : W \rightarrow K$ a linear map. Clearly

$$\ell \circ T : V \rightarrow K$$

is a linear map, b.c. both ℓ and T are linear. So, $T^*(\ell) := \ell \circ T \in V^*$. This shows that $T^* : W^* \rightarrow V^*$ indeed has as it's target V^* as claimed, see also the Image 4.2. We'll show now that T^* is a linear

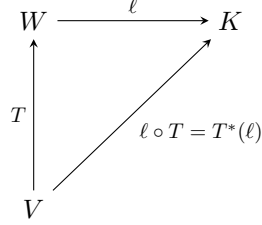


Figure 1: Map diagram of the dual map

map. Let $\ell \in W^*$, $\alpha \in K$. We have that $T^*(\alpha\ell) \in V^*$ is the map

$$\begin{aligned} T^*(\alpha\ell)(v) &= ((\alpha\ell) \circ T)(v) = (\alpha\ell)(Tv) = \alpha\ell(Tv) = \alpha(\ell \circ T)(v) = \alpha(T^*(\ell))(v) \\ \implies T^*(\alpha\ell) &= \alpha T^*(\ell). \end{aligned}$$

Now, let $\ell_1, \ell_2 \in W^*$. We have that $T^*(\ell_1 + \ell_2) \in V^*$ is the map

$$\begin{aligned} T^*(\ell_1 + \ell_2)(v) &= ((\ell_1 + \ell_2) \circ T)(v) = (\ell_1 \circ T + \ell_2 \circ T)(v) = (\ell_1 \circ T)(v) + (\ell_2 \circ T)(v) \\ &= (T^*(\ell_1))(v) + (T^*(\ell_2))(v) \\ \implies T^*(\ell_1 + \ell_2) &= T^*(\ell_1) + T^*(\ell_2). \end{aligned}$$

This shows that T^* is linear as claimed. □

Proposition 4.11: Composition of the Dual Map

Let U, V, W be vector spaces over K . Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps. Consider $T^* : W^* \rightarrow V^*$, $S^* : U^* \rightarrow W^*$, the dual maps. Then

$$(S \circ T)^* = T^* \circ S^*.$$

Lemma 4.12: Interpretation of the Matrix Transpose

Let $S : V \rightarrow W$ be a linear map between two finite dimensional vector spaces V and W over K . Let $\mathcal{B}$ be a basis for V and $\mathcal{C}$ be a basis for W . Consider $S^* : W^* \rightarrow V^*$ and the bases $\mathcal{C}^*, \mathcal{B}^*$ for W^*, V^* respectively. Then

$$[S^*]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([S]_{\mathcal{C}}^{\mathcal{B}})^T.$$

Proof. Write $\mathcal{B} = (v_1, \dots, v_n), \mathcal{C} = (w_1, \dots, w_m)$. Define $A := [S]_{\mathcal{C}}^{\mathcal{B}}, A = (a_{ij})$. A satisfies

$$Sv_j = \sum_{i=1}^m a_{ij}w_i.$$

Now, for $w_j^* \in \mathcal{C}^*$ we have by Lemma 4.8

$$\begin{aligned} S^*w_j^* &= w_j^* \circ S = \sum_{i=1}^n (w_j^* \circ S)(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^*(S(v_i)) \cdot v_i^* \\ &= \sum_{i=1}^n w_j^* \left(\sum_{k=1}^m a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*. \end{aligned}$$

Therefore, the coordinates of $S^*w_j^*$ in the basis $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ are

$$\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix},$$

which is the row $\#j$ in A , represented as a column, or equivalently the column $\#j$ in the matrix A^T . $\square$

4.3 Reflexivity

Let V be a vector space over K . We have V^* , but also $V^{**} := (V^*)^*$ by applying 'dual' one more time, to V^* . $(V^*)^*$ is sometimes called the bidual space of V . Is there any relation between $(V^*)^*$ and V .

Theorem 4.13: The Canonical Isomorphism between $(V^*)^*$ and V

Let v be a *finite* dimensional vector space over K . Then there exists a canonical isomorphism $\tau : V \rightarrow (V^*)^*$, given by

$$\forall v \in V, \tau(v) : V^* \rightarrow K$$

is the map

$$\tau(v)(\ell) := \ell(v) \quad \forall \ell \in V^*.$$

In other words,

$$V \ni v \xrightarrow{\tau} (V^* \ni \ell \mapsto \ell(v) \in K).$$

Remark 4.14. The map $\tau : V \rightarrow (V^*)^*$ exists and is defined in the same way as above regardless of V being finite or infinite dimensional. Moreover, τ is always injective. However, when V is infinite dimensional, τ fails to be surjective.

4.4 Naturality

The map $\tau : V \rightarrow (V^*)^*$ exists for every space V , and is defined in the same way. To distinguish τ 's for different spaces, denote it now by $\tau_V : V \rightarrow (V^*)^*$. Let V, W be two vector spaces over K . Then for every linear map $T : V \rightarrow W$ we have

$$\begin{array}{ccc}
V & \xrightarrow{\tau_V} & (V^*)^* \\
\downarrow T & \circlearrowleft & \downarrow (T^*)^* \\
W & \xrightarrow{\tau_W} & (W^*)^*
\end{array}$$

which we call naturality.

Proposition 4.15: Image of τ

Let V be a finite dimensional vector space over K , let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V . Let $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ be the dual basis and $(\mathcal{B}^*)^* = ((v_1^*)^*, \dots, (v_n^*)^*)$ the basis for $(V^*)^*$, which is dual to $\mathcal{B}^*$. Then $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$, i.e.,

$$\tau(v_i) = (v_i^*)^* \quad \forall 1 \leq i \leq n.$$

4.5 Direct Sums

Proposition 4.16: The Vector Space of the Cartesian Product

Let V, W be a vector space over K . Then the cartesian product $V \times W$ becomes a vector space over K if we endow it with the following operations:

- $\forall (v_1, w_1), (v_2, w_2) \in V \times W : (v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2).$
- $\forall \alpha \in K, (v, w) \in V \times W : \alpha \cdot (v, w) := (\alpha v, \alpha w) \in V \times W.$
- $0_{V \times W} := (0_V, 0_W).$

This new vector space $V \times W$ is usually denoted $V \oplus W$ and is called the direct sum of V and W . We write the elements of $V \oplus W$ as (v, w) or $v \oplus w$.

Remark 4.17. $V \oplus W$ comes together with the following **canonical maps**:

- 1) $i_V : V \rightarrow V \oplus W, \quad i_V(v) := (v, 0).$
- 2) $i_W : W \rightarrow V \oplus W, \quad i_W(w) := (0, w).$
- 3) $p_V : V \oplus W \rightarrow V, \quad p_V(v, w) := v.$
- 4) $p_W : V \oplus W \rightarrow W, \quad p_W(v, w) := w.$

The blue marked maps are canonical embeddings, while the orange marked maps are canonical projections. These maps are all linear. The maps p_V, p_W are surjective, and i_V, i_W are injective. Because of the maps i_V, i_W we can sometimes view V and W as subspaces of $V \oplus W$.

Proposition 4.18: The map φ

Let V be a vector space over K , $U \subseteq V$ a subspace and $U' \subseteq V$ another subspace which is a complement of U in V (recall Definition 1.42). Then the map

$$\varphi : U \oplus U' \rightarrow V, \quad \varphi(u, u') := u + u'$$

is a linear isomorphism.

Remark 4.19. When U' is a complement of U in V , then $U + U' = V$ and we also have a canonical isomorphism $U \oplus U' \xrightarrow{\cong} V$. But formally speaking $U \oplus U'$ is a different vector space than V .

4.5.1 Generalization to more Summands

Let $U_1, \dots, U_m$ be **finitely** many vector spaces over K . We can define $U_1 \oplus \dots \oplus U_m := U_1 \times \dots \times U_m$, with the coordinatewise/componentwise operations.

For **infinite** families $U_i, i \in \mathcal{I}$ one can define

$$\bigoplus_{i \in \mathcal{I}} U_i := \left\{ t : \mathcal{I} \rightarrow \bigcup_{i \in \mathcal{I}} U_i \mid t(i) \in U_i \quad \forall i \in \mathcal{I}, t(i) \neq 0 \text{ for at most finitely many } i\text{'s} \right\}.$$

4.6 Quotient Spaces

Let V be a vector space over K , and $U \subseteq V$ a subspace. We will define a new vector space over K , V/U , together with a linear map $\pi : V \rightarrow V/U$ s.t. π is surjective and $\ker(\pi) = U$. V/U will have some other good properties. V/U is called the **quotient of V by U** or V **modulo U** . π is called the **quotient map** or the projection onto the quotient space.

We begin with an equivalence relation on v . Define for $v_1, v_2 \in V$

$$v_1 \sim v_2 \quad :\Longleftrightarrow \quad v_1 - v_2 \in U.$$

We claim that $\sim$ defines an equivalence relation on V . Indeed

- **Reflexivity:** $v \sim v$, b.c. $v - v = 0 \in U$.
- **Symmetry:** If $v_1 \sim v_2 \implies v_2 \sim v_1$, b.c. $v_2 - v_1 = -(v_1 - v_2) \in U$.
- **Transitivity:** If $v_1 \sim v_2$ and $v_2 \sim v_3$, then $v_1 \sim v_3$, b.c. $v_1 - v_3 = \underbrace{(v_1 - v_2)}_{\in U} + \underbrace{(v_2 - v_3)}_{\in U} \in U$.

We denote the equivalence class of $v \in V$ by $[v]$. So

$$[v] = v + U = \{v + u \mid u \in U\},$$

or equivalently

$$[v] = \{w \in V \mid v \sim w\} = \{w \in V \mid v - w \in U\}.$$

Definition 4.20: Quotient Space

Let V be a vector space over K and $U \subseteq V$ a subspace. Define $V/U := V / \sim$ = the set of equivalence classes of the equivalence relation $\sim$. So, the elements of V/U are $[v]$, where $v \in V$. Alternatively, each element in V/U is a set of the form $v + U$, where $v \in V$. V/U is called the **quotient of V by U** (or modulo U).

We want to turn V/U into a vector space over K . To achieve this, we define:

- **Addition:** $\forall [v_1], [v_2] \in V/U, \quad [v_1] + [v_2] := [v_1 + v_2] \in V/U.$
- **Mult. by a scalar:** $\forall \alpha \in K, [v] \in V/U, \quad \alpha \cdot [v] := [\alpha v] \in V/U.$ (Alternatively written: $\alpha(v + U) := \alpha v + U.$)
- **The zero:** $0_{V/U} := [0_V].$ (Alternatively written: $0_{V/U} := 0_V + U.$)

Proposition 4.21: Well Definedness of The Operations

The operations above are well defined. Moreover, they give V/U the structure of a vector space over K .

We now define the map

$$\pi : V \rightarrow V/U, \quad v \mapsto [v] \in V/U.$$

Proposition 4.22: Properties of π

π is a linear map. Moreover, $\text{Im}(\pi) = V/U$ (i.e., π is surjective), and $\ker(\pi) = U$.

Proposition 4.23: Dimension of V/U

Suppose that V is finite dimensional. Then V/U is finite dimensional too, and $\dim V/U = \dim V - \dim U$.

Theorem 4.24: The Isomorphism Theorem

Let V, W be vector spaces over K and $T : V \rightarrow W$ a linear map. Define a new map $\bar{T} : V/\ker(T) \rightarrow \text{Im}(T)$, as follows. Let $x \in V/\ker(T)$. Choose a representative $v \in V$ for x , i.e., $x = [v]$. Define $\bar{T}(x) = \bar{T}([v]) := T(v)$. Then:

1. $\bar{T}$ is well defined, and is a linear map.
2. $\bar{T} : V/\ker(T) \rightarrow \text{Im}(T)$ is an isomorphism.
3. The Diagram in Figure 4.6 commutes, where $i : \text{Im}(T) \rightarrow W$ is the inclusion.

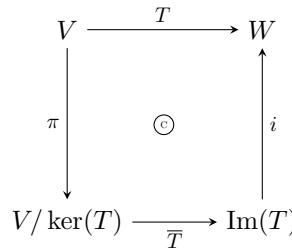


Figure 2: Map diagram of the map $\bar{T}$.

Proof. 1) $\bar{T}$ is well defined: Suppose $x = [v] = [v'] \in V/\ker(T)$, we need to show that $T(v) = T(v')$. Indeed, $[v] = [v']$ implies that $v - v' \in \ker(T)$. Therefore, $T(v - v') = 0$. Since T is linear we conclude that $T(v) = T(v')$.

$\bar{T}$ is linear: Let $\alpha \in K, [v] \in V/\ker(T)$. By the linearity of T , the definition of the operations on the quotient space $V/\ker(T)$ and the definition of $\bar{T}$, we have that

$$\bar{T}(\alpha[v]) = \bar{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha \bar{T}([v]).$$

Now, let $[v_1], [v_2] \in V/\ker(T)$. Then

$$\bar{T}([v_1] + [v_2]) = \bar{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \bar{T}([v_1]) + \bar{T}([v_2]).$$

2) $\bar{T}$ is injective: Let $x \in \ker(\bar{T})$. Write x as $x = [v]$ for some $v \in V$. Then

$$\underbrace{\bar{T}(x)}_{=0} = T(v).$$

Therefore, $v \in \ker(T)$. Thus, $x = [v] = [0_V] = 0_{V/\ker(T)}$.

$\bar{T}$ is surjective: Let $w \in \text{Im}(T)$. Then there exists a $v \in V$ s.t. $T(v) = w$. Therefore, $\bar{T}([v]) = T(v) = w$. This shows that $\text{Im}(T) \subseteq \text{Im}(\bar{T})$.

But by the definition of $\bar{T}$ we also have that $\text{Im}(\bar{T}) \subseteq \text{Im}(T)$.

3) The diagram commutes: Let $v \in V$ and $w \in \text{Im}(T) \subseteq W$ s.t. $T(v) = w$. Then by the definition of $\pi, \bar{T}$ and i , we have that

$$(i \circ \bar{T} \circ \pi)(v) = (i \circ \bar{T})([v]) = i(T(v)) = i(w) = w.$$

□

4.6.1 Universal Property of Quotient Spaces

Theorem 4.25: Universal Property of Quotient Spaces

Let V be a vector space over K and $U \subseteq V$ a subspace. The quotient space V/U has the following universal property:

For every space W and every linear map $T : V \rightarrow W$, for which $T(U) = 0$ (i.e., for which $U \subseteq \ker(T)$), there exists a **unique** linear map $T' : V/U \rightarrow W$ s.t. the diagram in Figure 4.6.1 commutes. Moreover, $\ker(T') = \ker(T)/U$.

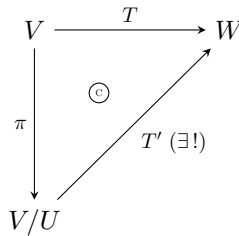


Figure 3: Map diagram of the map T' .

Jargon: Every linear map $T : V \rightarrow W$ that maps U to 0 factors through V/U (i.e., $T = T' \circ \pi$), and moreover in a unique way. We say that T **induces** a map $T' : V/U \rightarrow W$. We also say that T **descends** to a map $T' : V/U \rightarrow W$.

4.6.2 Quotient Spaces & Complements

Proposition 4.26: A Canonical Isomorphism to a Complement

Let V be a vector space over K and $U \subseteq V$ a subspace. Let $W \subseteq V$ be a complement of U . Then there exists a canonical isomorphism $Q : W \xrightarrow{\cong} V/U$, defined by

$$Q(w) := [w] \in V/U \quad \forall w \in W.$$

Remark 4.27. The isomorphism Q is canonical, only once W is specified. In general there is no canonical choice of a complement W to U . (In general there exist many choices of complements W to U , and there is no preferred choice.)

4.7 Relations to Dual Spaces

Proposition 4.28: Perpendicular Subspace

Let V be a vector space over K and $U \subseteq V$ a subspace. Define

$$U^\perp := \{\ell \in V^* \mid \ell|_U = 0\},$$

i.e., all the functionals on V that vanish on U . Then:

1. $U^\perp \subseteq V^*$ is a linear subspace.
2. There exists a **canonical** isomorphism $(V/U)^* \cong U^\perp$. If we assume that V is finite dimensional then there exists a **canonical** isomorphism $U^* \cong V^*/U^\perp$.

Proposition 4.29

Let $T : V \rightarrow W$ be a linear map. Then

$$(\text{Im}(T))^\perp = \ker(T^*).$$

5 Determinants

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(K).$$

There is a very simple criterion to check whether A is invertible or not.

Proposition 5.1: Criterion for Invertibility of a 2×2 Matrix

A is invertible if and only if $ad - bc \neq 0$. Moreover, if A is invertible, then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The number $ad - bc$ is $\det(A)$.

Definition 5.2: N-Linear Function

Let $D : M_{n \times n}(K) \rightarrow K$ be a function. We say that D is **n -linear** if for every $1 \leq i \leq n$, D is a linear function of the i -th row when the other $(n - 1)$ rows are held fixed.

We can think of $M_{n \times n}(K)$ as $K_{\text{row}}^n \times \dots \times K_{\text{row}}^n$ and if

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}$$

we write $D(A) = D(\alpha_1, \dots, \alpha_n)$. Saying that D is n -linear means that for all $1 \leq i \leq n$, we have that

1.

$$\begin{aligned} D(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) &= D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n) \\ \forall \alpha_1, \dots, \alpha_i, \alpha'_i, \dots, \alpha_n &\in K_{\text{row}}^n \end{aligned}$$

2.

$$\begin{aligned} D(\alpha_1, \dots, c \cdot \alpha_i, \dots, \alpha_n) &= c \cdot D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) \\ \forall c \in K, \alpha_1, \dots, \alpha_i, \dots, \alpha_n &\in K_{\text{row}}^n \end{aligned}$$

Lemma 5.3: Linear Combinations of n -linear Functions

Any linear combination of n -linear functions is an n -linear function again.

Definition 5.4: Alternating Functions

Let $D : M_{n \times n}(K) \rightarrow K$ be an n -linear function. We say that D is **alternating** if for every matrix $A \in M_{n \times n}(K)$ in which some two rows are equal we have that $D(A) = 0$.

Proposition 5.5

Let D be an n -linear function. Assume that D has the property that whenever A has two equal adjacent (aufeinanderfolgende) rows then $D(A) = 0$. Then

1. *D is alternating.*
2. *For every matrix B , if B' is obtained from B by interchanging any two of its rows, then $D(B') = -D(B)$.*

Definition 5.6: Determinant Function

A function $D : M_{n \times n}(K) \rightarrow K$ is called **determinant function** if D is n -linear, is alternating and $D(I) = 1$.

Definition 5.7: Row&Column-Deletion Matrix

Let $A \in M_{n \times n}(K)$, $n \geq 2$, $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by

$$A(i|j) \in M_{(n-1) \times (n-1)}(K)$$

the matrix obtained from A by deleting row $\#i$ and column $\#j$.

If D is an $(n-1)$ -linear function and $A \in M_{n \times n}(K)$ we denote

$$D_{ij}(A) := D(A(i|j)).$$

Notation: Let $A \in M_{m \times n}(K)$ and $1 \leq i \leq m$, $1 \leq j \leq n$. Denote the element at position (i, j) in the matrix A by $A(i, j)$ or A_{ij} .

Theorem 5.8: Entwicklungssatz

Let $n \geq 2$ and D an $(n-1)$ -linear function, which is alternating. Let $1 \leq j \leq n$. Define a function $E_j : M_{n \times n}(K) \rightarrow K$ by

$$E_j(A) := \sum_{i=1}^n (-1)^{i+j} A_{ij} D_{ij}(A).$$

Then E_j is an n -linear function and is alternating. If D is a determinant function then so is E_j .

Corollary 5.9: Existence of a Determinant Function

$\forall n \geq 1$, there exists at least one determinant function $M_{n \times n}(K) \rightarrow K$.

5.1 Uniqueness of the Determinant

Let $D : M_{n \times n}(K) \rightarrow K$ be an n -linear alternating function. Let

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix} \in M_{n \times n}(K).$$

Denote

$$I = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix} = \begin{pmatrix} - & \varepsilon_1 & - \\ & \vdots & \\ - & \varepsilon_n & - \end{pmatrix}.$$

(So, $\varepsilon_1, \dots, \varepsilon_n$ is the standard basis of K_{row}^n .) Clearly,

$$\alpha_i = \sum_{j=1}^n A(i, j) \varepsilon_j \quad \forall 1 \leq i \leq n.$$

Therefore,

$$\begin{aligned} D(A) &= D(\alpha_1, \dots, \alpha_n) = D\left(\sum_{j=1}^n A(1, j) \varepsilon_j, \alpha_2, \dots, \alpha_n\right) = \sum_{j=1}^n A(1, j) \cdot D(\varepsilon_j, \alpha_2, \dots, \alpha_n) \\ &= \sum_{j=1}^n A(1, j) \cdot D\left(\varepsilon_j, \sum_{k=1}^n A(2, k) \varepsilon_k, \dots, \alpha_n\right) = \sum_{j,k=1}^n A(1, j) \cdot A(2, k) \cdot D(\varepsilon_j, \varepsilon_k, \dots, \alpha_n). \end{aligned}$$

Applying these steps to every row we get that

$$D(A) = \sum_{k_1, \dots, k_n=1}^n A(1, k_1) \cdot A(2, k_2) \cdot \dots \cdot A(n, k_n) \cdot D(\varepsilon_{k_1}, \varepsilon_{k_2}, \dots, \varepsilon_{k_n}), \quad (5.1)$$

where each of the indices $k_1, \dots, k_n$ runs between 1 and n . So far we haven't used the fact that D is alternating. Now, since D is alternating $D(\varepsilon_{k_1}, \dots, \varepsilon_{k_n}) = 0$, whenever two of the indices $k_1, \dots, k_n$ are equal. So, we are interested only in ordered sequences of indices $(k_1, \dots, k_n)$ in which $k_i \in \{1, \dots, n\} \quad \forall i$ and $k_i \neq k_j \quad \forall i \neq j$. Such a sequence is called a permutation of $\{1, \dots, n\}$, or a permutation of degree n .

Definition 5.10: Permutation

A **permutation** is a bijective map $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$.

So Alternatively, we can think of a permutation of degree n as a bijective function $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. Then the sequence $(k_1, \dots, k_n) = (\sigma 1, \dots, \sigma n)$ is a permutation as defined earlier. (Notation: $\sigma(i) := \sigma i$.) In other words, a permutation σ of degree n is a re-ordering of a list with n elements $(1, \dots, n) \xrightarrow{\sigma} (\sigma 1, \dots, \sigma n)$.

We can rewrite Equation (5.1) as

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdot \dots \cdot A(n, \sigma n) \cdot D(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}),$$

where S_n is the set of all permutations of degree n .

5.1.1 Important Thing About Permutations

Let σ be a permutation of degree n . Then one can pass from $(1, \dots, n)$ to $(\sigma 1, \dots, \sigma n)$ by performing a finite sequence of interchanges of pairs, called transpositions.

Definition 5.11: Transposition

A **transposition** is a permutation $\theta : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, where $\exists i, j \in \{1, \dots, n\}, i \neq j$ s.t. $\theta(i) = j, \theta(j) = i$ and $\theta(k) = k \quad \forall k \neq i, j$.

Now, every permutation $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ can be written as a composition of transpositions

$$\sigma = \theta_1 \circ \dots \circ \theta_m, \quad (5.2)$$

where $\theta_1, \dots, \theta_m$ are transpositions. But these θ_i 's and m are not unique. In other words, the same σ can be written as in Equation (5.2) in many different ways

$$\sigma = \theta_1 \circ \dots \circ \theta_m = \theta'_1 \circ \dots \circ \theta'_{m'}.$$

Why Can σ be written as in (5.2). Start with $(1, \dots, n)$ and if $\sigma 1 \neq 1$ then interchange 1 and

$\sigma 1$. We now have $(\sigma 1, \dots, 1, \dots, n)$. If $\sigma 2 \neq 2$, then interchange 1 and $\sigma 2$; otherwise proceed to 3, etc. But of course there are other ways to do it, e.g., $(3, 2, 1)$ can be obtained in 3 steps: $(1, 2, 3) \rightarrow (2, 1, 3) \rightarrow (2, 3, 1) \rightarrow (3, 2, 1)$. But also in one step $(1, 2, 3) \rightarrow (3, 2, 1)$.

Definition 5.12: Sign of a Permutation

Let σ be a permutation and let $\theta_1, \dots, \theta_m$ be transpositions s.t. $\sigma = \theta_1 \circ \dots \circ \theta_m$. The permutation σ is called **even** if m is even, and is called **odd** if m is odd. The number $(-1)^m \in \{-1, 1\}$ is called the **sign** of σ , i.e., $\text{sgn}(\sigma) := (-1)^m$.

Lemma 5.13: Well Definedness of the Sign of a Permutation

If $(\sigma 1, \dots, \sigma n)$ is obtained from $(1, \dots, n)$ in two different ways, one time by a sequence of m transpositions, and another time by a sequence of m' transpositions, then m' and m have the same parity, i.e., $(-1)^{m'} = (-1)^m$.

(So the parity of m depends only on σ and not on the particular way we performed a sequence of transpositions $(1, \dots, n) \rightarrow \dots \rightarrow (\sigma 1, \dots, \sigma n)$.)

Proof. We know that determinant functions $M_{n \times n}(K) \rightarrow K$ exists for every $n \geq 1$. Fix a determinant function $E : M_{n \times n}(K) \rightarrow K$. Consider $E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n})$. If $(\sigma 1, \dots, \sigma n)$ can be obtained from $(1, \dots, n)$ by a sequence of m transpositions, then the matrix

$$\begin{pmatrix} - & \varepsilon_1 & - \\ & \vdots & \\ - & \varepsilon_n & - \end{pmatrix}$$

can be obtained from

$$I = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$$

by a sequence of m interchanges of pairs of rows. Therefore,

$$E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}) = (-1)^m E(\varepsilon_1, \dots, \varepsilon_n) = (-1)^m E(I) = (-1)^m.$$

But this applies also to m' . So,

$$E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}) = (-1)^{m'} E(\varepsilon_1, \dots, \varepsilon_n) = (-1)^{m'} E(I) = (-1)^{m'}.$$

So we have that

$$(-1)^m = (-1)^{m'}.$$

□

5.1.2 Back to our $D(A)$

We have seen that if D is n -linear and alternating then

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdot \dots \cdot A(n, \sigma n) \cdot D(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}).$$

But

$$D(\varepsilon_{\sigma 1}, \varepsilon_{\sigma n}) = (\operatorname{sgn} \sigma) \cdot D(\varepsilon_1, \dots, \varepsilon_n) = (\operatorname{sgn} \sigma) \cdot D(I) \quad (5.3)$$

Theorem 5.14: Uniqueness of the Determinant Function

$\forall n \geq 1$, there exists a unique determinant function $M_{n \times n}(K) \rightarrow K$. We will denote it by $\det$. It is given by the formula

$$\det(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdots A(n, \sigma n).$$

Corollary 5.15

For every n -linear and alternating function $D : M_{n \times n}(K) \rightarrow K$ we have that

$$D(A) = \det(A) \cdot D(I) \quad \forall A \in M_{n \times n}(K).$$

5.1.3 A Bit more on Permutations

Denote by S_n the set of permutations $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. S_n has the structure of a group:

- We can compose permutations, i.e., if $\sigma, \tau : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, then $\sigma \cdot \tau := \sigma \circ \tau$, where $\sigma \circ \tau : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$.
- The identity $id : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ is the neutral element $1 \in S_n$.
- The inverse σ^{-1} of σ , is given by the inverse of the function σ (recall that σ is bijective, so σ^{-1} does exist).

Also, note that $|S_n| = n!$

Lemma 5.16: Sign of the Composition of Permutations

$\forall \sigma, \tau \in S_n$ we have

$$\operatorname{sgn}(\sigma \cdot \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau).$$

Theorem 5.17: The Determinant of the Product of Two Matrices

$\forall A, B \in M_{n \times n}(K)$ we have that

$$\det(A \cdot B) = \det(A) \cdot \det(B).$$

Proof. Fix a matrix $B \in M_{n \times n}(K)$, and consider a function $D : M_{n \times n}(K) \rightarrow K$ defined by

$$D(A) := \det(A \cdot B).$$

Claim: D is n -linear and alternating.

Proof of Claim: Write

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}.$$

If we view α_i as a $1 \times n$ matrix, then $\alpha_i B$ is also $1 \times n$, and in fact

$$A \cdot B = \begin{pmatrix} - & \alpha_1 B & - \\ & \vdots & \\ - & \alpha_n B & - \end{pmatrix}.$$

So,

$$D(A) = \det(\alpha_1 B, \dots, \alpha_n B).$$

Clearly, for every two row vectors α_i, α'_i we have that $(\alpha_i + \alpha'_i)B = \alpha_i B + \alpha'_i B$ and

$$\forall c \in K : (c \cdot \alpha_i)B = c \cdot (\alpha_i B).$$

Since $\det$ is n -linear it follows that D is also n -linear.

If $\alpha_i = \alpha_j$, then $\alpha_i B = \alpha_j B$. Therefore,

$$D(A) = \det(\alpha_1 B, \dots, \alpha_n B) = 0.$$

This proves the claim.

We continue with the proof of the Theorem. Since we found that D is n -linear and alternating, by Corollary 5.15, we have that

$$D(A) = \det(A) \cdot D(I).$$

But $D(I) = \det(I \cdot B) = \det(B)$. Therefore,

$$D(A) = \det(A \cdot B) = \det(A) \cdot \det(B). \quad \square$$

Corollary 5.18: Equivalent Condition for The Invertibility of a Matrix

If A is invertible then $\det(A) \neq 0$. Moreover,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

5.2 More on Determinants

Theorem 5.19: Determinant of a Transposed Matrix

$$\forall A \in M_{n \times n}(K) : \det(A^T) = \det(A).$$

Corollary 5.20

If $D : M_{n \times n}(K) \rightarrow K$ is an alternating n -linear function (of the rows of a matrix) then D is also an alternating and n -linear function of the columns of a matrix. In fact, $D(A^T) = D(A) \quad \forall A \in M_{n \times n}(K)$. In particular, this holds also for $D = \det$.

Theorem 5.21: Determinant after Row/Column Operations

1. If B is obtained from A by adding a multiple of one row of A to another row (or by doing the analogous operation on columns), then $\det(B) = \det(A)$.
2. If $A \xrightarrow{R_i \leftrightarrow R_j} B$, $i \neq j$, (or the analogous operation on columns) then $\det(B) = -\det(A)$.
3. If $A \xrightarrow{c \cdot R_i \rightarrow R_i} B$, then $\det(B) = c \cdot \det(A)$.

5.2.1 Determinants of Block Matrices

Consider a block matrix M of the type

$$M = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix},$$

where $A \in M_{r \times r}(K)$, $C \in M_{s \times s}(K)$, $B \in M_{r \times s}(K)$.

Proposition 5.22: Determinant of a Block Matrix

Let M be as above. Then

$$\det(M) = \det(A) \cdot \det(C).$$

Corollary 5.23: Entwicklungssatz with Cofactors

$\forall A \in M_{n \times n}(K)$ we have the following n different formulas for $\det(A)$, i.e.,

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} A_{ij} \det(A(i|j)), \quad j = 1, \dots, n.$$

The scalars $c_{ij} := (-1)^{i+j} \det(A(i|j))$ are called the i, j **cofactor** of A . We have

$$\det(A) = \sum_{i=1}^n A_{ij} \cdot c_{ij}. \tag{5.4}$$

Claim: If we replace A_{ij} in Equation (5.4) by A_{ik} , where k is fixed, we always get 0. In other words, $\forall A \in M_{n \times n}(K)$, $\forall j$ we have

$$\sum_{i=1}^n A_{ik} \cdot c_{ij} = \begin{cases} \det(A) & j = k \\ 0 & j \neq k \end{cases}.$$

Proof of Claim: Let $B \in M_{n \times n}(K)$ be the matrix obtained from A by replacing column j of A by column k of A . So, in B column j equals column k . Clearly, $\det(B) = 0$ since B has two equal columns ($\iff B^T$ has two equal rows).

$$\implies 0 = \det(B) = \sum_{i=1}^n (-1)^{i+j} B_{ij} \det(B(i|j)) = \sum_{i=1}^n = \sum_{i=1}^n A_{ik} \det(A(i|j)) = \sum_{i=1}^n A_{ik} \cdot c_{ij}.$$

This proves the claim.

So, $\forall A \in M_{n \times n}(K)$, $\forall j, k$ we have

$$\sum_{i=1}^n A_{ik} \cdot c_{ij} = \delta_{jk} \cdot \det(A). \quad (5.5)$$

Definition 5.24: Classical Adjoint of A

The matrix $(c_{ij})^T$ is called the **classical adjoint of A** , i.e.,

$$\text{adj}(A) \in M_{n \times n}(K), \quad (\text{adj}(A))_{ij} = c_{ji} = (-1)^{i+j} \cdot \det(A(j|i)).$$

The Equation (5.5) says that

$$(\text{adj } A) \cdot A = \det(A) \cdot I.$$

Indeed,

$$\begin{aligned} \det(A) \cdot \delta_{jk} &= \sum_{i=1}^n c_{ij} \cdot A_{ik} = \sum_{i=1}^n (\text{adj}(A))_{ji} \cdot A_{ik} \\ \implies \det(A) \cdot I &= \text{adj}(A) \cdot A. \end{aligned}$$

Theorem 5.25: Formula for the inverse of A

Let $A \in M_{n \times n}(K)$. Then A is invertible if and only if $\det(A) \neq 0$. When A is invertible we have

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

5.3 Cramer's Rule

Let $A \in M_{n \times n}(K)$. We would like a formula to solve the system of equations $Ax = b$, where

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

are the unknowns and

$$b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n$$

is given. Assume $Ax = b$. Then

$$\begin{aligned} \text{adj } A \cdot Ax &= \text{adj } A \cdot b \\ \implies \det(A) \cdot x &= \text{adj } A \cdot b \\ \implies \forall 1 \leq j \leq n : \det(A) \cdot x_j &= \sum_{i=1}^n (\text{adj } A)_{ji} \cdot b_i = \sum_{i=1}^n (-1)^{i+j} b_i \det(A(i|j)), \end{aligned}$$

where the last expression is the determinant of the matrix obtained from A by replacing column j by b .

Theorem 5.26: Cramer's Rule

Let $A \in M_{n \times n}(K)$ with $\det(A) \neq 0$. Let $b \in K^n$. Then the system $Ax = b$ has a unique solution, and it is given by the formula

$$x_j = \frac{\det(A(j, b))}{\det(A)} \quad \forall 1 \leq j \leq n,$$

where $A(j, b)$ is the matrix obtained from A by replacing column j by b .

5.4 Determinants of Endomorphisms

Let $A \in M_{n \times n}(K)$ and $P \in GL_n(K)$, then

$$\det(P^{-1}AP) = \det(P^{-1}) \cdot \det(A) \cdot \det(P) = \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) = \det(A).$$

In other words, similar matrices have the same determinant.

Let V be a finite dimensional vector space over K . Put $n := \dim V$. Let $T : V \rightarrow V$ be a linear map. Pick a basis $\mathcal{B}$ for V . We have a matrix $[T]_{\mathcal{B}}^{\mathcal{B}} \in M_{n \times n}(K)$. Consider $\det[T]_{\mathcal{B}}^{\mathcal{B}} \in K$.

Question: Does $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ depend on the basis $\mathcal{B}$?

Answer: Let $\mathcal{B}'$ be another basis for V . Denote by $P := [id_V]_{\mathcal{B}}^{\mathcal{B}'}$ the transition matrix between the basis $\mathcal{B}'$ and $\mathcal{B}$. Recall that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'},$$

and we also have $[id_V]_{\mathcal{B}'}^{\mathcal{B}} = ([id_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1}$. So,

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = P^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot P. \quad \implies \quad \det[T]_{\mathcal{B}'}^{\mathcal{B}'} = \det[T]_{\mathcal{B}}^{\mathcal{B}}.$$

Recall also that if $T, S : V \rightarrow V$ are two linear maps then $[S \circ T]_{\mathcal{B}}^{\mathcal{B}} = [S]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}$. Recall also that if $T : V \rightarrow V$ is an isomorphism then $[T^{-1}]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^{-1}$. Finally, $[id_V]_{\mathcal{B}}^{\mathcal{B}} = I$.

Corollary 5.27: Well Definedness of the Determinant of an Endomorphism

Let V be a finite dimensional vector space over K . Let $T : V \rightarrow V$ be a linear map. Then $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ does **not** depend on the choice of basis $\mathcal{B}$ of V . We denote this scalar by $\det(T) \in K$. The function

$$\det : \text{End}(V) \rightarrow K, \quad T \mapsto \det(T)$$

has the following properties:

1. $\det(S \circ T) = \det(S) \cdot \det(T) \quad \forall S, T \in \text{End}(V)$.
2. T is an isomorphism if and only if $\det(T) \neq 0$. If T is an isomorphism then

$$\det(T^{-1}) = \frac{1}{\det(T)}.$$

3. $\det(id_V) = 1$.
4. $\det(0) = 0 \in K$.

Remark 5.28. In contrast to $T : V \rightarrow V$, if we take $T : V \rightarrow W$, and $\mathcal{B}, \mathcal{C}$ bases of V, W , then $\det[T]_{\mathcal{C}}^{\mathcal{B}}$ **depends** on $\mathcal{B}, \mathcal{C}$ (unless $W = V$ and $\mathcal{C} = \mathcal{B}$).

6 Eigenvalues & Eigenvectors

Definition 6.1: Eigenvalues & Eigenvectors

Let V be a vector space over K and $T : V \rightarrow V$ a linear map (an endomorphism of V). We say that $\lambda \in K$ is an **eigenvalue** of T if $\exists v \in V \setminus \{0\}$ s.t.

$$Tv = \lambda v.$$

Any vector $v \in V \setminus \{0\}$ with the property that $Tv = \lambda v$ is called an **eigenvector** of T for the eigenvalue λ .

Definition 6.2: Diagonalizability

Let V be a finite dimensional vector space over K and $T : V \rightarrow V$ a linear map. We say that T is diagonalizable if there exists a basis $\mathcal{B}$ of V s.t.

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}. \quad (6.1)$$

Lemma 6.3: Equivalent Condition for Diagonalizability

T is diagonalizable if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ consisting of eigenvectors of T (i.e., $\forall i, Tv_i = \lambda_i v_i$ for some $\lambda_i \in K$). Moreover, if for some basis $\mathcal{B} = (v_1, \dots, v_n)$ the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape as in Equation (6.1), then $\forall i, \lambda_i$ is an eigenvalue of T and v_i is an eigenvector of λ_i .

Remark 6.4. We saw that $\forall A \in M_{n \times n}(K)$ we have a linear map $T_A : K^n \rightarrow K^n$, $T_A(v) := Av$. We say that $A \in M_{n \times n}(K)$ is diagonalizable if the linear map T_A is diagonalizable.

Proposition 6.5: Linear Independence of Eigenvectors of Distinct Eigenvalues

Let V be a vector space over K and $T : V \rightarrow V$ a linear map. Suppose that $\lambda_1, \dots, \lambda_n \in K$ are eigenvalues of T which are pairwise distinct (i.e., $\lambda_i \neq \lambda_j \ \forall i \neq j$). Let $v_1, \dots, v_n$ be eigenvectors for $\lambda_1, \dots, \lambda_n$ respectively. Then $v_1, \dots, v_n$ are linearly independent.

Proof. Induction on n .

$n = 1$: v_1 is an eigenvector for λ_1 . By definition of $v_1 \neq 0 \implies v_1$ is linearly independent.

Induction step: Let $n \geq 1$ and suppose the proposition holds for any list of n eigenvalues and eigenvectors. We'll prove now that the proposition holds also for a list of $n + 1$ eigenvalues and eigenvectors. Let $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ be pairwise distinct eigenvalues of T , and $v_1, \dots, v_n, v_{n+1}$ a given list of eigenvectors of $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$, respectively. We need to prove that $v_1, \dots, v_n, v_{n+1}$ are linearly independent.

Assume that

$$a_1 v_1 + \dots + a_n v_n + a_{n+1} v_{n+1} = 0. \quad (6.2)$$

We need to show that $a_1 = \dots = a_n = a_{n+1} = 0$. Apply T to Equation (6.2). Then we have that

$$a_1 T v_1 + \dots, a_n T v_n + a_{n+1} T v_{n+1} = 0.$$

But $\forall i : T v_i = \lambda_i v_i$. Therefore,

$$a_1 \lambda_1 v_1 + \dots, a_n \lambda_n v_n + a_{n+1} \lambda_{n+1} v_{n+1} = 0. \quad (6.3)$$

Consider now Equation (6.3) - $\lambda_{n+1} \cdot (6.2)$, i.e.,

$$a_1 (\lambda_1 - \lambda_{n+1}) v_1 + \dots, a_n (\lambda_n - \lambda_{n+1}) v_n + a_{n+1} \underbrace{(\lambda_{n+1} - \lambda_{n+1})}_{=0} v_{n+1} = 0.$$

By the induction hypothesis $v_1, \dots, v_n$ are linearly independent. Therefore,

$$a_1 (\lambda_1 - \lambda_{n+1}) = \dots, a_n (\lambda_n - \lambda_{n+1}) = 0.$$

By assumption $\lambda_i \neq \lambda_{n+1} \quad \forall 1 \leq i \leq n$. So we have that $a_1 = \dots = a_n = 0$. Going back to Equation (6.2) we get that $a_{n+1} \cdot v_{n+1} = 0$. But by assumption we also have that $v_{n+1} \neq 0$. Therefore, $a_{n+1} = 0$. $\square$

6.1 The Characteristic Polynomial

Proposition 6.6: Equivalent Condition for Eigenvalue

*Let V be a vector space over K and $T : V \rightarrow V$ a linear map. Then $\lambda \in K$ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot \text{id}$ is **not** injective.*

Definition 6.7: The Characteristic Polynomial

Let V be a finite dimensional vector space over K and $T : V \rightarrow V$ a linear map. We define the **characteristic polynomial of T** as

$$p_T(x) := \det(T - x \cdot \text{id}_V).$$

We define the characteristic polynomial of a matrix $A \in M_{n \times n}(K)$ in the same way, i.e.,

$$p_A(x) := \det(A - x \cdot I_n).$$

Definition 6.8: Eigenspace

Let V be a vector space over K and $T \in \text{End}(V)$. Let $\lambda \in K$ be an eigenvalue of T . The **eigenspace** corresponding to λ is defined as

$$\text{Eig}_T(\lambda) := \{v \in V \mid T v = \lambda v\} = \ker(T - \lambda \text{id}_V) = \{v \in V \mid v \text{ is an eigenvector of } \lambda\} \cup \{0\}.$$

Clearly, since $\text{Eig}_T(\lambda)$ is the kernel of a linear map, $\text{Eig}_T(\lambda) \subseteq V$ is a linear subspace of V .

Lemma 6.9: Properties of the Characteristic Polynomial

Let $A \in M_{n \times n}(K)$. Then:

1. $p_A(x)$ is a polynomial of degree n , with leading coefficient $(-1)^n$, i.e., $p_A(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x + c_0$, with $c_i \in K \quad \forall i$, and $c_n = (-1)^n$.
2. $p_{A^T}(x) = p_A(x)$.
3. If A has upper triangular form as in Definition 3.12, then

$$p_A(x) = (\lambda_1 - x) \cdot \dots \cdot (\lambda_n - x).$$

4. If

$$A = \begin{pmatrix} B_1 & * \\ 0 & B_2 \end{pmatrix}$$

is a block matrix, then

$$p_A(x) = p_{B_1}(x) \cdot p_{B_2}(x).$$

5. If A and B are similar matrices, i.e., $B = P^{-1}AP$ for some $P \in GL_n(K)$, then

$$p_A(x) = p_B(x).$$

Remark 6.10. If V is an n -dimensional vector space over K and $T \in \text{End}(V)$, then $p_T(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$.

Definition 6.11: The Trace

Let $A \in M_{n \times n}(K)$. Define the **trace** (Spur) of A as

$$\text{Tr}(A) := \sum_{i=1}^n A_{ii} \in K,$$

i.e., the sum of the entries along the diagonal of A . If V is a finite dimensional vector space over K and $T \in \text{End}(V)$, we define the trace of T as

$$\text{Tr}(T) := \text{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}),$$

where $\mathcal{B}$ is some basis for V . We'll see soon that $\text{Tr}(T)$ is well defined, i.e., it does not depend on the choice of $\mathcal{B}$.

Lemma 6.12: Properties of the Trace

1.

$$\forall A, B \in M_{n \times n}(K) : \operatorname{Tr}(A \cdot B) = \operatorname{Tr}(B \cdot A).$$

2. If A and B are similar matrices, then

$$\operatorname{Tr}(A) = \operatorname{Tr}(B).$$

In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are two bases for V , then

$$\operatorname{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \operatorname{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'}).$$

Thus, $\operatorname{Tr}(T)$ is well defined.

Proof. 1):

$$\begin{aligned} \operatorname{Tr}(A \cdot B) &= \sum_{i=1}^n (A \cdot B)_{ii} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} \cdot B_{ji} \\ &= \sum_{j=1}^n \sum_{i=1}^n A_{ij} \cdot B_{ji} = \sum_{j=1}^n \sum_{i=1}^n B_{ji} A_{ij} = \sum_{j=1}^n (B \cdot A)_{jj} \\ &= \operatorname{Tr}(B \cdot A). \end{aligned}$$

2): If $B = P^{-1}AP$ for some $P \in GL_n(K)$, then by 1) we have that

$$\operatorname{Tr}(B) = \operatorname{Tr}(P^{-1}AP) = \operatorname{Tr}(P^{-1} \cdot (AP)) = \operatorname{Tr}((AP) \cdot P^{-1}) = \operatorname{Tr}(A).$$

Regarding the claim about $\operatorname{Tr}(T)$, recall that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'} \quad (6.4)$$

and $[id_V]_{\mathcal{B}}^{\mathcal{B}} = ([id_V]_{\mathcal{B}'}^{\mathcal{B}'})^{-1}$. So, by Equation (6.4) we have that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [T]_{\mathcal{B}}^{\mathcal{B}}. \quad \square$$

Lemma 6.13: Coefficients of the Characteristic Polynomial

Let $A \in M_{n \times n}(K)$ and $p_A(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x + c_0$ its characteristic polynomial. We've already seen that $c_n = (-1)^n$. We also have that

$$c_{n-1} = (-1)^{n-1} \cdot \operatorname{Tr}(A), \quad c_0 = \det(A).$$

In other words,

$$p_A(x) = (-1)^n x^n + (-1)^{n-1} x^{n-1} \operatorname{Tr}(A) + c_{n-2} x^{n-2} + \dots + c_1 x + \det(A).$$

Definition 6.14: Direct Sum

Let V be a vector space over K and $U_1, \dots, U_r \subseteq V$ subspaces. Denote $W := U_1 + \dots + U_r \subseteq V$. We say that W is a **direct sum** of $U_1, \dots, U_r$ (or that $U_1, \dots, U_r$ is a direct sum) if every $w \in W$ has a **unique** representation as $w = u_1 + \dots + u_r$ with $u_i \in U_i \quad \forall i$.

Proposition 6.15: Eigenspaces are a Direct Sum

Let V be a vector space over K and $T \in \text{End}(V)$. Let $\lambda_1, \dots, \lambda_r \in K$ be pairwise distinct eigenvalues of T . Then

$$\text{Eig}_T(\lambda_1) + \dots + \text{Eig}_T(\lambda_r)$$

is a direct sum.

Corollary 6.16: Equivalent Condition for Diagonalizability of an Endomorphism

Assume V is finite dimensional, $T \in \text{End}(V)$ and let $\lambda_1, \dots, \lambda_r \in K$ be all the eigenvalues of T . Assume additionally that $\lambda_1, \dots, \lambda_r$ are pairwise distinct. Then T is diagonalizable if and only if

$$\sum_{i=1}^r \dim \text{Eig}_T(\lambda_i) = \dim V.$$

Lemma 6.17: Decomposition of the Characteristic Polynomial

Let $A \in M_{n \times n}(K)$ (or $T \in \text{End}(V)$) be diagonalizable. Then $p_A(x)$ decomposes as a product of linear factors, i.e., degree 1

$$p_A(x) = (\lambda_1 - x) \cdot \dots \cdot (\lambda_n - x),$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (but there might be repetitions among the λ_i 's).

6.1.1 A Quick Digression about Polynomials

Let $0 \neq p(x) \in K[x]$, $\lambda \in K$. By dividing $p(x)$ by $(x - \lambda)$ we can write

$$p(x) = (x - \lambda) \cdot q(x) + R,$$

where $\deg q(x) = \deg p(x) - 1$ and $R \in K$. But if we substitute $x = \lambda$, we obtain

$$p(\lambda) = 0 \cdot q(\lambda) + R = R \implies p(x) = (x - \lambda) \cdot q(x) + p(\lambda).$$

Conclusion: $p(\lambda) = 0 \iff p(x) = (x - \lambda) \cdot q(x)$ for some $q(x) \in K[x]$. We say that λ is a zero of $p(x)$ (Nullstelle) or a root of $p(x)$.

Sometimes λ is also a zero of $q(x)$, so $q(x) = (x - \lambda) \cdot h(x)$. Therefore, $p(x) = (x - \lambda)^2 \cdot h(x)$, etc.

Definition 6.18: Multiplicity of Zeroes

Let $0 \neq p(x) \in K[x]$, $\lambda \in K$. We say that λ is a **zero of multiplicity** $m \in \mathbb{N}_{>0}$ if

$$p(x) = (x - \lambda)^m \cdot g(x),$$

where $g(x) \in K[x]$ and $g(\lambda) \neq 0$.

If $m \geq 2$ we say λ is a **multiple zero** of $p(x)$.

If $m = 1$ we say λ is a **simple zero** of $p(x)$.

6.2 Geometric & Algebraic Multiplicities of Eigenvalues**Definition 6.19: Geometric & Algebraic Multiplicities**

Let V be a vector space over K and $T \in \text{End}(V)$. Let $\lambda \in K$ be an eigenvalue of T . We define the **geometric multiplicity** of λ to be

$$m_g(T; \lambda) := \dim \text{Eig}_T(\lambda).$$

Assume now, in addition, that V is finite dimensional. We define the **algebraic multiplicity** of λ to be the multiplicity of the zero λ in the characteristic polynomial $p_T(x)$ of T . We denote it by $m_A(T; \lambda)$.

Proposition 6.20: Inequality Regarding Geometric & Algebraic Multiplicity

$$m_g(T; \lambda) \leq m_A(T; \lambda).$$

Proof. Let $v_1, \dots, v_k$ be a basis for $\text{Eig}_T(\lambda)$. We extend this basis to a basis $\mathcal{B} = (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$ of V . Then we have

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} | & & | & & | & & | \\ [Tv_1]_{\mathcal{B}} & \dots & [Tv_n]_{\mathcal{B}} & [Tw_{k+1}]_{\mathcal{B}} & \dots & [Tw_n]_{\mathcal{B}} \\ | & & | & & | & & | \end{pmatrix} = \begin{pmatrix} \lambda & \dots & 0 & * \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & \lambda & * \\ 0 & \dots & 0 & C \end{pmatrix},$$

where $C \in M_{(n-k) \times (n-k)}(K)$. Therefore,

$$p_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - xI_n) = (\lambda - x)^k \cdot \det(C - xI_{n-k}) = (\lambda - x)^k \cdot p_C(x) = (-1)^k \cdot (x - \lambda)^k \cdot p_C(x).$$

this shows that the multiplicity of λ as a zero of $p_T(x)$ is at least k . In other words,

$$m_A(T; \lambda) \geq k = m_g(T; \lambda).$$

□

6.3 Diagonalizability

Theorem 6.21: Equivalent Statements for Diagonalizability

Let V be an n -dimensional vector space over K and $T \in \text{End}(V)$. The following statements are equivalent:

1. T is diagonalizable.
2. There exists a basis for V whose elements are eigenvectors of T .
3. The characteristic polynomial $p_T(x)$ splits into a product of linear factors, and for every eigenvalue of T the geometric and algebraic multiplicities are equal.

4.

$$\sum_{i=1}^k \dim \text{Eig}_T(\lambda_i) = \dim V,$$

where $\lambda_1, \dots, \lambda_k \in K$ are the eigenvalues of T .

5.

$$V \cong \bigoplus_{i=1}^k \text{Eig}_T(\lambda_i).$$

Corollary 6.22

If $p_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity 1, then T is diagonalizable.

Corollary 6.23

If $p_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity 1, then T is diagonalizable.

6.4 Trigonalization

Definition 6.24: Trigonalizability

We say that $T \in \text{End}(V)$ is **trigonalizable** if there exists a basis $\mathcal{B}$ for V s.t. $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular, i.e.,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & * \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}.$$

Theorem 6.25: Equivalent Condition for Trigonalizability

Let V be a finite dimensional vector space over K and $T \in \text{End}(V)$. Then T is trigonalizable if and only if $p_T(x)$ splits as a product of linear factors, i.e.,

$$p_T(x) = (\lambda_1 - x)^{l_1} \cdots (\lambda_k - x)^{l_k}.$$

Theorem 6.26: Fundamental Theorem of Algebra

Every polynomial $p(x) \in \mathbb{C}[x]$ with complex coefficients, splits as a product of linear factors and a constant, i.e.,

$$p(x) = c \cdot (x - a_1)^{n_1} \cdot \dots \cdot (x - a_r)^{n_r},$$

with $c \in \mathbb{C}$, $r \geq 0$, $a_i \in \mathbb{C}$, $n_j \in \mathbb{N}_{>0}$.

Corollary 6.27: Trigonalizability over $\mathbb{C}$

Let V be a finite dimensional vector space over $\mathbb{C}$. Then every $T \in \text{End}(V)$ is trigonalizable.

6.5 Minimal Polynomial & Cayley-Hamilton Theorem**Definition 6.28: Matrix Polynomial**

Let $A \in M_{n \times n}(K)$. Let $g(x) = a_d x^d + \dots + a_1 x + a_0 \in K[x]$ be a polynomial. We can define $g(A) \in M_{n \times n}(K)$, as

$$g(A) := a_d A^d + \dots + a_1 A + a_0 I_n.$$

Remark 6.29. $\forall A \in M_{n \times n}(K) \exists g \in K[x]$ s.t. $g(A) = 0$. Indeed, consider

$$I_n, A, A^2, \dots, A^r \in M_{n \times n}(K). \quad (6.5)$$

Since $M_{n \times n}(K)$ is a vector space of dimension n^2 , then if we take $r > n^2$ then the elements in (6.5) cannot be linearly independent. Therefore, $\exists a_0, a_1, \dots, a_r \in K$ s.t.

$$a_0 I_n + a_1 A + \dots + a_r A^r = 0.$$

A similar thing applies to $T \in \text{End}(V)$, where V is a finite dimensional vector space over K .
 $(T^k = \underbrace{T \circ \dots \circ T}_{\times k}, g(T) := a_d T^d + \dots + a_1 T + a_0 \text{id}_V.)$

Definition 6.30: Minimal Polynomial

Let V be a vector space over K and $T \in \text{End}(V)$. A **minimal polynomial** for T is a polynomial $g(x) \in K[x]$ of the minimal possible degree s.t. $g(T) = 0$.

Theorem 6.31: Cayley-Hamilton

1. Let $A \in M_{n \times n}(K)$. Then

$$p_A(A) = 0.$$

2. Let V be a finite dimensional vector space over K , and $T \in \text{End}(V)$. Then

$$p_T(T) = 0.$$

The proof of this Theorem may be important for the exam but will be skipped in this Document. The proof can be found in the lecture notes.

7 Euclidean & Hermitian (unitary) Spaces

At last we endow our spaces with a geometric structure. We work here with vector spaces V over $K = \mathbb{R}$ or $K = \mathbb{C}$. Over $\mathbb{R}$ we'll have Euclidean spaces. Over $\mathbb{C}$ we'll have Hermitian (a.k.a unitary) spaces.

Definition 7.1: Scalar Product

Let V be a vector space over $\mathbb{R}$. A **scalar product** (a.k.a inner product) is a function $V \times V \rightarrow \mathbb{R}$, $(v, w) \mapsto \langle v, w \rangle$ that has the following properties:

1. Linearity in the first variable:

$$\begin{aligned}\langle v_1 + v_2, w \rangle &= \langle v_1, w \rangle + \langle v_2, w \rangle & \forall v_1, v_2, w \in V \\ \langle \alpha v, w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{R}, v, w \in V.\end{aligned}$$

2. Linearity in the second variable:

$$\begin{aligned}\langle v, w_1 + w_2 \rangle &= \langle v, w_1 \rangle + \langle v, w_2 \rangle & \forall v, w_1, w_2 \in V \\ \langle v, \alpha w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{R}, v, w \in V.\end{aligned}$$

3. Symmetry:

$$\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V.$$

4. Positivity:

$$\langle v, v \rangle > 0 \quad \forall v \in V \setminus \{0\}.$$

We call $(V, \langle \cdot, \cdot \rangle)$ an **Euclidean space**, or a space endowed with a scalar product.

Definition 7.2: Hermitian Product

Let V be a vector space over $\mathbb{C}$. A **Hermitian product** (or inner product) on V is a function $V \times V \rightarrow \mathbb{C}$, $(v, w) \mapsto \langle v, w \rangle$ with the following properties:

1. Linearity in the first variable:

$$\begin{aligned}\langle v_1 + v_2, w \rangle &= \langle v_1, w \rangle + \langle v_2, w \rangle & \forall v_1, v_2, w \in V \\ \langle \alpha v, w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{C}, v, w \in V.\end{aligned}$$

2. Sesquilinearity (complex anti-linearity) in the second variable:

$$\begin{aligned}\langle v, w_1 + w_2 \rangle &= \langle v, w_1 \rangle + \langle v, w_2 \rangle & \forall v, w_1, w_2 \in V \\ \langle v, \alpha w \rangle &= \bar{\alpha} \langle v, w \rangle & \forall \alpha \in \mathbb{C}, v, w \in V.\end{aligned}$$

3. Hermitian property:

$$\langle v, w \rangle = \overline{\langle w, v \rangle} \quad \forall v, w \in V.$$

4. Positivity:

$$\langle v, v \rangle > 0 \quad \forall v \in V.$$

We call $(V, \langle \cdot, \cdot \rangle)$ a **Hermitian (or unitary) space**.

Lemma 7.3: Properties of Euclidean/Hermitian Spaces

Let V be an Euclidean/Hermitian vector space. Then:

1. $\forall v \in V : \langle 0, v \rangle = \langle v, 0 \rangle = 0$.
2. Let $w \in V$. If $\langle v, w \rangle = 0 \quad \forall v \in V$, then $w = 0$.
3. Let $w_1, w_2 \in V$. If $\langle v, w_1 \rangle = \langle v, w_2 \rangle \quad \forall v \in V$, then $w_1 = w_2$.

Definition 7.4: Some Definitions to Euclidean/Hermitian Spaces

1. Let V be an Euclidean/Hermitian vector space. $\forall v \in V$, define $\|v\| := \sqrt{\langle v, v \rangle} \in \mathbb{R}_{\geq 0}$. $\|\cdot\|$ is called the **norm** on V induced by $\langle \cdot, \cdot \rangle$.

2. A vector $u \in V$ is called a **unit vector** if $\|u\| = 1$. Note that if $0 \neq v \in V$ is any vector then

$$\frac{v}{\|v\|}$$

is a unit vector (which is positively proportional to v , so it points in the same direction as v). We call $\frac{v}{\|v\|}$ the normalization of v .

3. For $v, w \in V$ we define the **distance** between v and w by

$$d(v, w) := \|v - w\|.$$

4. Two vectors $u, v \in V$ are called **orthogonal** (or perpendicular one to the other) if $\langle u, v \rangle = 0$. In this case we sometimes write $u \perp v$.

5. A subset $S \subseteq V$ is called an **orthogonal system** if $0 \notin S$ and for every two distinct elements $u, v \in S$ we have $u \perp v$.

6. An orthogonal system $S \subseteq V$ is called **orthonormal** if $\forall v \in S$ we have $\|v\| = 1$.

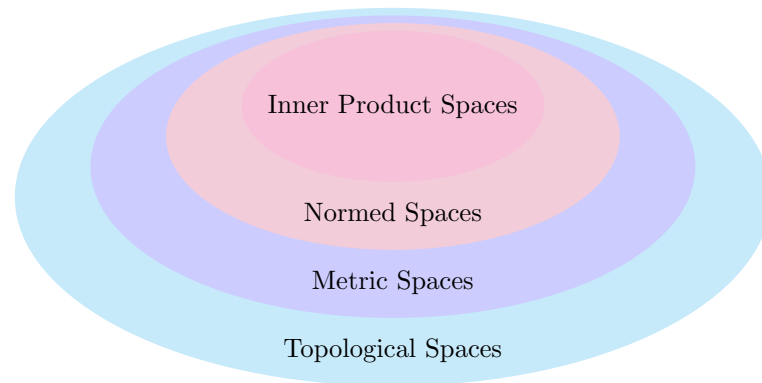


Figure 4: Overview of the generality of different spaces

Theorem 7.5: Pythagoras Theorem

Let V be an Euclidean/Hermitian vector space and $u, v \in V$ with $u \perp v$. Then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Lemma 7.6: Induced Norm

Let V be an Euclidean space. Then

$$\forall u, v \in V : \langle u, v \rangle = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2).$$

This means that $\|\cdot\|$ determines $\langle \cdot, \cdot \rangle$.

If V is an Hermitian space, we have that

$$\forall u, v \in V : \frac{1}{2}(\langle u, v \rangle + \overline{\langle u, v \rangle}) = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2).$$

Warning: There exist norms that do **not** come from scalar products.

Definition 7.7: Complex Conjugated Matrix/Adjoint Matrix/Hermitian Matrix

1. Let $A \in M_{n \times m}(\mathbb{C})$. We write $\overline{A}$ for the matrix whose entry at i, j is $\overline{(A_{ij})}$, i.e.,

$$(\overline{A})_{ij} := \overline{(A_{ij})}.$$

2. Let $A \in M_{n \times n}(\mathbb{C})$. Define

$$A^* := \overline{A}^T.$$

A^* is called the **adjoint matrix of A** . (Has nothing to do with the classical adjoint $\text{adj } A$).

3. $A \in M_{n \times n}(\mathbb{C})$ is called **Hermitian** if $A^* = A$ ($\iff A^T = \overline{A}$).

Important Example: Let $A \in M_{n \times n}(\mathbb{R})$. Define

$$\langle \cdot, \cdot \rangle_A : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}, \quad \langle v, w \rangle_A = v^T A w.$$

We can quickly see that $\langle \cdot, \cdot \rangle_A$ is bilinear. Assume now that A is symmetric. Then $\langle v, w \rangle_A = \langle w, v \rangle_A \quad \forall v, w \in V$. Indeed,

$$\langle w, v \rangle_A = w^T A v = w^T A^T v = (v^T A w)^T = (\langle v, w \rangle_A)^T = \langle v, w \rangle_A,$$

where we've used that a scalar can be viewed as a 1×1 matrix.

Is $\langle \cdot, \cdot \rangle_A$ in general a scalar product?

In general it is not, e.g., for

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

we have that

$$\left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle_A = -1.$$

But for a scalar product we need that $\langle v, v \rangle > 0 \quad \forall v \neq 0$.

Definition 7.8: Positive Definite

A symmetric matrix $A \in M_{n \times n}(\mathbb{R})$ is called **positive definite** if

$$v^T A v > 0 \quad \forall v \in \mathbb{R}^n \setminus \{0\}.$$

Proposition 7.9: Scalar Product with Matrices

Let $A \in M_{n \times n}(\mathbb{R})$. Then $\langle \cdot, \cdot \rangle_A$ is a scalar product if and only if A is positive definite.

The situation for Hermitian products is similar.

Definition 7.10: Positive Definite for Hermitian Matrices

Let $A \in M_{n \times n}(\mathbb{C})$ be a Hermitian matrix. We say that A is **positive definite** if

$$v^T A \bar{v} > 0 \quad \forall v \in \mathbb{C}^n \setminus \{0\}.$$

Note that $v^T A \bar{v} \in \mathbb{R}$, b.c.

$$\overline{(v^T A \bar{v})} = \bar{v}^T \bar{A} v = (\bar{v}^T \bar{A} v)^T = v^T \bar{A}^T \bar{v} = v^T A \bar{v}.$$

Given a positive definite complex matrix A we can define a Hermitian product on $\mathbb{C}^n$, i.e.,

$$\langle v, w \rangle_A = v^T A \bar{w}.$$

Proposition 7.11: Hermitian Product with Matrices

Let $A \in M_{n \times n}(\mathbb{C})$. Then $\langle \cdot, \cdot \rangle_A$ is a Hermitian product if and only if A is positive definite.

7.1 Norms & Angles**Definition 7.12: Norm**

Let V be a vector space over K , where $K = \mathbb{R}$ or $\mathbb{C}$. A **norm** on V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ s.t.:

1. Triangle inequality:

$$\|u + v\| \leq \|u\| + \|v\| \quad \forall u, v \in V.$$

2. Homogeneity:

$$\|\alpha v\| = |\alpha| \cdot \|v\| \quad \forall \alpha \in K, v \in V.$$

3. Non-degeneracy:

$$\|v\| = 0 \implies v = 0.$$

Theorem 7.13: Cauchy-Schwarz Inequality

Let $(V, \langle \cdot, \cdot \rangle)$ be a vector space endowed with an inner product $\langle \cdot, \cdot \rangle$. Then

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\| \quad \forall u, v \in V,$$

where equality holds if and only if u and v are linearly dependent.

Corollary 7.14: Induced Norm

Let V be a vector space and $\langle \cdot, \cdot \rangle$ an inner product on V , then

$$V \ni v \mapsto \|v\| := \sqrt{\langle v, v \rangle}$$

is a norm on V .

Definition 7.15: Angle

Let $(V, \langle \cdot, \cdot \rangle)$ be an Euclidean space. Let $u, v \in V$ be two non-zero vectors. By the Cauchy-Schwarz inequality 7.13, we have that

$$-1 \leq \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} \leq 1.$$

We define the **angle** between u and v to be the unique $\alpha \in [0, \pi]$ s.t.

$$\cos \alpha = \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} \left(= \left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle \right)$$

7.2 The Gram-Schmidt Orthogonalization Process**Theorem 7.16: Properties of Orthogonal Systems**

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space.

1. If $S \subseteq V$ is an orthogonal system then S is linearly independent.
2. If $v_1, \dots, v_n$ is an orthogonal system and $v \in \text{Sp}(v_1, \dots, v_n)$ and write $v = a_1 v_1 + \dots + a_n v_n$, for $a_j \in K$. Then the coefficients a_j are given by

$$a_j = \frac{\langle v, v_j \rangle}{\langle v_j, v_j \rangle} \quad \forall 1 \leq j \leq n.$$

Corollary 7.17: Matrix for an Orthonormal Basis

Let V, W be a vector space over $K = \mathbb{R}$ or $\mathbb{C}$ and let $\langle \cdot, \cdot \rangle$ be an inner product on W . Let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V and $\mathcal{C} = (w_1, \dots, w_m)$ be an orthonormal basis for W . Let $T : V \rightarrow W$ be a linear map. Write $[T]_{\mathcal{C}}^{\mathcal{B}} = (a_{ij})$. Then

$$a_{ij} = \langle T v_j, w_i \rangle \quad \forall i, j.$$

(If $\mathcal{C}$ is only a orthogonal basis then $a_{ij} = \langle T v_j, w_i \rangle / \langle w_i, w_i \rangle$).

So having an ortho-gonal/normal basis for our spaces is useful!

Theorem 7.18: Gram-Schmidt Orthogonalization Process

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$ and let $n = \dim V$. Let $(v_1, \dots, v_n)$ be a basis for V . Define a new list of vectors $w_1, \dots, w_n$ by induction, as follows

$$w_1 := v_1,$$

$$w_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} \cdot w_i \quad \text{for } j = 2, \dots, n.$$

Then:

1. $(w_1, \dots, w_n)$ is an orthogonal basis for V .
2. $(\frac{w_1}{\|w_1\|}, \dots, \frac{w_n}{\|w_n\|})$ is an orthonormal basis for V .
3. $\forall 1 \leq j \leq n$ we have $\text{Sp}(v_1, \dots, v_j) = \text{Sp}(w_1, \dots, w_j) = \text{Sp}(\frac{w_1}{\|w_1\|}, \dots, \frac{w_j}{\|w_j\|})$.

Corollary 7.19: Existence of an Orthonormal Basis for Finite Dimensional Spaces

Every finite dimensional inner product space has an orthonormal basis.

7.3 The Orthogonal Complement

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$.

Definition 7.20: Orthogonal Complement

Let $\emptyset \neq S \subseteq V$ be a subset. We define the **orthogonal complement of S** as

$$S^\perp := \{v \in V \mid v \perp s \quad \forall s \in S\} = \{v \in V \mid \langle v, s \rangle = 0 \quad \forall s \in S\}.$$

When $S = \{v\}$ we write $v^\perp := \{v\}^\perp$.

Lemma 7.21: The Orthogonal Complement is a Subspace

Let $\emptyset \neq S \subseteq V$ be a subset. Then $S^\perp$ is a linear subspace.

Proof. 1) $0 \in S^\perp$ b.c. $\langle 0, s \rangle = 0$ for every $s \in S$. Furthermore, if $\alpha, \beta \in K$ and $v, w \in S^\perp$ then for every $s \in S$ we have that

$$\langle \alpha v + \beta w, s \rangle = \underbrace{\alpha \langle v, s \rangle}_{=0} + \underbrace{\beta \langle w, s \rangle}_{=0} = 0.$$

Therefore, $\alpha v + \beta w \in S^\perp$. □

Lemma 7.22: Further Properties of the Orthogonal Complement

Let $\emptyset \neq S, T \subseteq V$. Then:

1. $0^\perp = V$, $V^\perp = \{0\}$.
2. $S \cap S^\perp$ is either $\emptyset$ or equal to $\{0\}$.
3. If $S \subseteq T$ then $T^\perp \subseteq S^\perp$.
4. $(\text{Sp}(S))^\perp = S^\perp$.
5. $S \subseteq (S^\perp)^\perp$.

Theorem 7.23: The Orthogonal Complement is a Complement

Let V be an inner product space (possibly of infinite dimension), and $U \subseteq V$ a **finite dimensional** subspace. Then $U^\perp$ is a complement of U in V (, i.e., $U + U^\perp = V$ and $U \cap U^\perp = \{0\}$).

Proof. By Lemma 7.22 2., we have that $U \cap U^\perp = \{0\}$, since $U \cap U^\perp \neq \emptyset$. So it's enough to show that $U + U^\perp = V$.

Let $r := \dim U$ and pick an orthonormal basis $(e_1, \dots, e_r)$ for U , which we can do by the Gram-Schmidt Orthogonalization Process 7.18. Let $v \in V$ be arbitrary. Define

$$P_U(v) := \langle v, e_1 \rangle e_1 + \dots + \langle v, e_r \rangle e_r \in U.$$

We have that

$$v = \underbrace{P_U(v)}_{\in U} + (v - P_U(v)). \quad (7.1)$$

We claim that $v - P_U(v) \in U^\perp$. Indeed,

$$\langle v - P_U(v), e_i \rangle = \langle v, e_i \rangle - \sum_{k=1}^r \langle v, e_k \rangle \cdot \langle e_k, e_i \rangle = \langle v, e_i \rangle - \langle v, e_i \rangle = 0.$$

This holds for every $1 \leq i \leq r$, hence

$$v - P_U(v) \perp \text{Sp}(e_1, \dots, e_r) = U,$$

i.e., $v - P_U(v) \in U^\perp$. Going back to Equation (7.1) we see that

$$v = \underbrace{P_U(v)}_{\in U} + \underbrace{(v - P_U(v))}_{\in U^\perp}.$$

Since $v \in V$ was arbitrary, we have that $U + U^\perp = V$. □

Remark 7.24. If $U \subseteq V$ is a finite dimensional subspace, then since $U^\perp$ is a complement of U in V , we have a canonical isomorphism $V \cong U \oplus U^\perp$.

Definition 7.25: The Orthogonal Projection onto a Subspace

Let V be an inner product space and $U \subseteq V$ a finite dimensional subspace. We have seen that $U^\perp$ is a complement of U in V ($\implies \forall v \in V, \exists! u \in U, w \in U^\perp$ s.t. $v = u + w$).

Define a map $P_U : V \rightarrow U$ as follows:

$$\forall v \in V, \exists! u \in U, w \in U^\perp \text{ s.t. } v = u + w.$$

We define

$$P_U(v) := u.$$

P_U is called **the orthogonal projection onto U** .

Proposition 7.26: Properties of P_U

P_U has the following properties:

1. P_U is a linear map.
2. $\text{Im } P_U = U, \text{ ker } P_U = U^\perp$.
3. $\forall v \in V : v - P_U(v) \in U^\perp$, i.e., $P_U + P_{U^\perp} = \text{id}_V$.
4. $\forall u \in U : P_U(u) = u$.
5. If $e_1, \dots, e_r$ is any orthonormal basis for U , then

$$P_U(v) = \sum_{i=1}^r \langle v, e_i \rangle e_i.$$

Corollary 7.27: The Complement of the Complement of a Subspace

Let V be an inner product space and $U \subseteq V$ a finite dimensional subspace. Then

$$(U^\perp)^\perp = U.$$

Corollary 7.28: Dimension of a Subspace and its Complement

If V is a finite dimensional inner product space and $U \subseteq V$ is a subspace, then

$$\dim U + \dim U^\perp = \dim V.$$

7.4 Orthogonal & Unitary Matrices

Definition 7.29: Orthogonal & Unitary Matrices

1. A matrix $A \in M_{n \times n}(\mathbb{R})$ is called **orthogonal** if its columns form an orthonormal basis in $\mathbb{R}^n$, w.r.t. the standard scalar product of $\mathbb{R}^n$. We denote the set of orthogonal $n \times n$ matrices by $O(n) \subseteq M_{n \times n}(\mathbb{R})$.
2. A matrix $A \in M_{n \times n}(\mathbb{C})$ is called **unitary** if its columns form an orthonormal basis in $\mathbb{C}^n$, w.r.t. the standard hermitian product on $\mathbb{C}^n$. We denote the set of all unitary $n \times n$ matrices by $U(n) \subseteq M_{n \times n}(\mathbb{C})$.

Lemma 7.30: Equivalent Conditions for Orthogonal/Unitary Matrices

- $A \in M_{n \times n}(\mathbb{R})$ is orthogonal $\iff A^T A = I_n \iff A A^T = I_n \iff A^{-1} = A^T$.
- $A \in M_{n \times n}(\mathbb{C})$ is unitary $\iff A^* A = I_n \iff A A^* = I_n \iff A^{-1} = A^*$.

Corollary 7.31

1. Let $A \in M_{n \times n}(\mathbb{R})$. Then $A \in O(n) \iff$ the rows of A form an orthonormal basis for $\mathbb{R}^n$.
2. Let $A \in M_{n \times n}(\mathbb{C})$. Then $A \in U(n) \iff$ the rows of A form an orthonormal basis for $\mathbb{C}^n$.

7.5 QR-Decomposition

Theorem 7.32: QR-Decomposition

1. Let $A \in GL_n(\mathbb{R})$. Then $\exists Q \in O(n)$ and an upper triangular matrix $R \in M_{n \times n}(\mathbb{R})$ s.t.

$$A = QR.$$

2. Let $A \in GL_n(\mathbb{C})$. Then $\exists Q \in U(n)$ and an upper triangular matrix $R \in M_{n \times n}(\mathbb{C})$ s.t.

$$A = QR.$$

Proof. We'll prove 1 & 2 together.

Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

with $v_k \in K_{\text{col}}^n$, where $K = \mathbb{R}$ or $\mathbb{C}$. Since $A \in GL_n(K)$, $v_1, \dots, v_n$ form a basis for K_{col}^n . Apply the Gram-Schmidt orthogonalization process 7.18 to $v_1, \dots, v_n$ and obtain an orthonormal basis $e_1, \dots, e_n$ for K_{col}^n . Recall that $\forall 1 \leq j \leq n : \text{Sp}(v_1, \dots, v_j) = \text{Sp}(e_1, \dots, e_j)$. By Theorem 7.16 we also have that

$$v_j = \sum_{i=1}^j \langle v_j, e_i \rangle e_i \quad \forall 1 \leq j \leq n.$$

Therefore, we can write

$$\underbrace{\begin{pmatrix} | & & | & & | \\ v_1 & \dots & v_j & \dots & v_n \\ | & & | & & | \end{pmatrix}}_{=A} = \underbrace{\begin{pmatrix} | & & | & & | \\ e_1 & \dots & e_j & \dots & e_n \\ | & & | & & | \end{pmatrix}}_{=Q} \cdot \underbrace{\begin{pmatrix} \langle v_1, e_1 \rangle & \langle v_2, e_1 \rangle & \dots & \langle v_j, e_1 \rangle & \dots & \langle v_n, e_1 \rangle \\ 0 & \langle v_2, e_2 \rangle & \dots & \langle v_j, e_2 \rangle & \dots & \langle v_n, e_2 \rangle \\ \vdots & \ddots & \ddots & \vdots & & \vdots \\ \vdots & & \ddots & \langle v_j, e_j \rangle & \dots & \langle v_n, e_j \rangle \\ \vdots & & & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 & \langle v_n, e_n \rangle \end{pmatrix}}_{=R},$$

as desired. $\square$

Extension of QR-Decomposition for all matrices $A \in M_{n \times n}(K)$.

Theorem 7.33: Extension of QR-Decomposition

Let $A \in M_{n \times n}(K)$, where $K = \mathbb{R}$ or $\mathbb{C}$ and let $r := \text{rank}(A)$. Then $\exists Q \in O(n)/U(n)$ and

$$R = \begin{pmatrix} C & * \\ 0 & 0 \end{pmatrix} \in M_{n \times n}(K),$$

where $C \in M_{r \times r}(K)$ is upper triangular, s.t.

$$A = QR.$$

8 Dual Spaces & Inner Products

Let V be a vector space over K (any field). Recall that we have the dual space $V^* := \text{Hom}(V, K) =$ the space of linear maps $\varphi : V \rightarrow K$ (functionals). V^* is also a vector space over K .

Consider now an inner product space $(V, \langle \cdot, \cdot \rangle)$ over $K = \mathbb{R}$ or $\mathbb{C}$. Let $u \in V$. Define

$$\varphi_u : V \rightarrow K, \quad V \ni v \mapsto \langle v, u \rangle \in K.$$

φ_u is linear since, v is the first argument in the scalar/hermitian product.

Theorem 8.1: Canonical 'Almost' Isomorphism between V and V^*

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space over $K = \mathbb{R}$ or $\mathbb{C}$. Let $\varphi \in V^*$. Then there exists a **unique** $u \in V$ s.t. $\varphi = \varphi_u$, i.e., $\varphi(v) = \varphi_u(v) \quad \forall v \in V$.

In fact, for $K = \mathbb{R}$ the map $\Phi : V \rightarrow V^*$, given by $\Phi(u) := \varphi_u$ is an isomorphism.

For $K = \mathbb{C}$, this map is injective and surjective but only $\mathbb{C}$ -anti linear, i.e.,

$$\begin{aligned} \Phi(u_1 + u_2) &= \Phi(u_1) + \Phi(u_2) \quad \forall u_1, u_2 \in V \\ \Phi(cu) &= \bar{c} \cdot \Phi(u) \quad \forall c \in \mathbb{C}, u \in V. \end{aligned}$$

So, at least for $K = \mathbb{R}$, a scalar product on V gives us a canonical isomorphism $V \cong V^*$. (But this isomorphism depends on the scalar product!)

8.1 The Adjoint Map

(Everything in this Subsection should be learned by heart.) Let V, W be vector spaces over K . Let $T : V \rightarrow W$ be a linear map. We have seen in Subsection 4.2 that T induces a linear map (the dual map)

$$T^* : W^* \rightarrow V^*, \quad T^*(\varphi) := \varphi \circ T.$$

Assume now that $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ are inner product spaces over $K = \mathbb{R}$ or $\mathbb{C}$. Assume also that $\dim V < \infty$. Given T as above, we can define $T' : W \rightarrow V$ using our 'canonical' identifications $W \cong W^*$, $V \cong V^*$. (No problem for $K = \mathbb{R}$, but we have to be careful with $K = \mathbb{C}$.)

$$\begin{array}{ccc} W^* & \xrightarrow{T^*} & V^* \\ \uparrow \Phi_W & & \downarrow \Phi_V^{-1} \\ W & \xrightarrow{\quad T' \quad} & V \end{array}$$

i.e. $T' := \Phi_V^{-1} \circ T^* \circ \Phi_W$. We've also used that $\dim V < \infty$, b.c. then Φ_V is bijective.

What properties does T' have? Note that T' is uniquely determined by

$$\begin{aligned} \Phi_V \circ T' &= T^* \circ \Phi_W &\iff \forall w \in W : \Phi_V \circ T'(w) &= T^* \circ \Phi_W(w) \\ &&\iff \varphi_{T'(w)} &= T^* \circ \varphi_w \quad (\in V^*) \\ &&\iff \forall v \in V : \varphi_{T'(w)}(v) &= (T^* \circ \varphi_w)(v) \\ &&\iff \langle v, T'(w) \rangle_V &= \varphi_w(T(v)) = \langle T(v), w \rangle_W. \end{aligned}$$

In other words,

$$\langle Tv, w \rangle_W = \langle v, T'w \rangle_V \quad \forall v \in V, w \in W.$$

Claim: T' is a linear map.

Proof. For $K = \mathbb{R}$, Φ_W, Φ_V and T^* are linear. Therefore, T' is linear too.

For $K = \mathbb{C}$, Φ_W, Φ_V are additive but only $\mathbb{C}$ -anti linear. Therefore, Φ_V^{-1} is also additive and $\mathbb{C}$ -anti linear. Indeed, let $y_1, y_2 \in V^*$. Since Φ_V is bijective, we can find unique $x_1, x_2 \in V$ s.t. $\Phi_V(x_1) = y_1$ and $\Phi_V(x_2) = y_2$. Since Φ_V is additive we have that

$$\Phi_V^{-1}(y_1 + y_2) = \Phi_V^{-1}(\Phi_V(x_1) + \Phi_V(x_2)) = \Phi_V^{-1}(\Phi_V(x_1 + x_2)) = x_1 + x_2 = \Phi_V^{-1}(y_1) + \Phi_V^{-1}(y_2).$$

Similarly, let $c \in \mathbb{C}, y \in V^*$. We again can find a unique $x \in V$ s.t. $\Phi_V(x) = y$. Since Φ_V is $\mathbb{C}$ -anti linear we have that

$$\Phi_V^{-1}(cy) = \Phi_V^{-1}(c \cdot \Phi_V(x)) = \Phi_V^{-1}(\Phi_V(\bar{c}x)) = \bar{c} \cdot x = \bar{c} \cdot \Phi_V^{-1}(y).$$

So, we have shown that Φ_V^{-1} is additive and $\mathbb{C}$ -anti linear as well.

Now, we return to the proof of the claim. We recall that T^* is linear. Let $w_1, w_2 \in W$ be arbitrary. Since T^* is in particular additive and $T' = \Phi_V^{-1} \circ T^* \circ \Phi_W$, we can conclude that T' is also additive.

Now, let $c \in \mathbb{C}, w \in W$. Since T^* is linear and Φ_V^{-1}, Φ_W are $\mathbb{C}$ -anti linear we have that

$$\begin{aligned} T'(cw) &= \Phi_V^{-1} \circ T^* \circ \Phi_W(cw) = \Phi_V^{-1} \left(T^* (\bar{c} \cdot \Phi_W(w)) \right) = \Phi_V^{-1} \left(\bar{c} \cdot T^*(\Phi_W(w)) \right) \\ &= c \cdot \Phi_V^{-1} \left(T^*(\Phi_W(w)) \right) = c \cdot T'(w). \end{aligned}$$

This proves the claim. $\square$

Notation: The linear map $T' : W \rightarrow V$ is denoted many times by $T^* : W \rightarrow V$. This makes sense b.c. of the 'canonical' identifications. But we have to assume additionally that $\dim W < \infty$. The map $T^* : W \rightarrow V$ is called the **adjoint** of T .

8.2 Basic Properties of Adjoint Maps

Lemma 8.2: Properties of Adjoint Maps

Let V, W, U be finite dimensional inner product spaces over $K = \mathbb{R}$ or $\mathbb{C}$. Let $S, T : V \rightarrow W$ and $R : W \rightarrow U$ be linear maps. Then:

1. T^* is a linear map.
2. $(S + T)^* = S^* + T^*$.
3. $\forall \lambda \in K : (\lambda T)^* = \bar{\lambda} T^*$.
4. $(T^*)^* = T$.
5. $(id_V)^* = id_V$.
6. $(R \circ T)^* = T^* \circ R^*$.

Proof. Let $y_1, y_2 \in V$. We'll repeatedly use that if $\forall x \in V : \langle x, y_1 \rangle = \langle x, y_2 \rangle$, then $y_1 = y_2$. This is because

$$\begin{aligned} \forall x \in V : \langle x, y_1 \rangle &= \langle x, y_2 \rangle &\implies & \forall x \in V : \langle x, y_1 - y_2 \rangle = 0 \\ \implies y_1 - y_2 &= 0 &\implies & y_1 = y_2, \end{aligned}$$

where we've used 2. of Lemma 7.3

- 1) Has already been proven in Subsection 8.1.
- 2) Let $w \in W, v \in V$. By the defining properties of T^*, S^* we have that

$$\langle v, (S + T)^* w \rangle = \langle (S + T)v, w \rangle = \langle Sv, w \rangle + \langle Tv, w \rangle = \langle v, S^* w \rangle + \langle v, T^* w \rangle = \langle v, (S^* + T^*)w \rangle.$$

This holds for every $v \in V$. Therefore, we have that

$$(S + T)^* w = (S^* + T^*)w.$$

Since, this holds for every $w \in W$, we conclude that $(S + T)^* = (S^* + T^*)$.

- 3) Let $\lambda \in K, v \in V$ and $w \in W$. Then we have that

$$\begin{aligned} \langle v, (\lambda T)^* w \rangle &= \langle (\lambda T)v, w \rangle = \lambda \langle Tv, w \rangle = \lambda \langle v, T^* w \rangle \\ &= \langle v, \bar{\lambda} T^* w \rangle. \end{aligned}$$

Since this holds for every $v \in V$, we get that

$$(\lambda T)^* w = \bar{\lambda} T^* w.$$

Since this holds for every $w \in W$, we conclude that $(\lambda T)^* = \bar{\lambda} T^*$.

4) Let $v \in V$, $w \in W$. Then we have that

$$\langle w, (T^*)^* v \rangle = \langle T^* w, v \rangle = \overline{\langle v, T^* w \rangle} = \overline{\langle T v, w \rangle} = \langle w, T v \rangle.$$

This holds for every $w \in W$. Therefore, we have that $(T^*)^* v = T v$. And since this holds for every $v \in V$, we get that $(T^*)^* = T$.

5) Let $v \in V$. Then we have that

$$\langle v, (id_V)^* v \rangle = \langle id_V v, v \rangle = \langle v, v \rangle = \langle v, id_V v \rangle.$$

Since this holds for every $v \in V$ we conclude that

$$(id_V)^* v = id_V v \quad \implies \quad (id_V)^* = id_V.$$

6) Let $v \in V$, $u \in U$. Then we have that

$$\begin{aligned} \langle v, (R \circ T)^* u \rangle &= \langle (R \circ T) v, u \rangle = \langle R(T(v)), u \rangle = \langle T v, R^* u \rangle \\ &= \langle v, T^* (R^* (u)) \rangle = \langle v, T^* \circ R^* u \rangle. \end{aligned}$$

This holds for every $v \in V$. Therefore,

$$(R \circ T)^* u = T^* \circ R^* u.$$

Since this holds for every $u \in U$, we conclude that $(R \circ T)^* = T^* \circ R^*$. □

Lemma 8.3: Kernel & Image of Adjoint Maps

Let V, W be finite dimensional inner product spaces. Let $T : V \rightarrow W$ and $T^* : W \rightarrow V$ be the adjoint of T . Then

1. $\ker T^* = (\operatorname{Im} T)^\perp$.
2. $\operatorname{Im} T^* = (\ker T)^\perp$.
3. $\ker T = (\operatorname{Im} T^*)^\perp$.
4. $\operatorname{Im} T = (\ker T^*)^\perp$.

8.3 Matrix Representation of Adjoint Maps

Proposition 8.4: Matrix Representation of Adjoint Maps

Let V, W be finite dimensional inner product spaces and $TV \rightarrow W$ a linear map. Let $\mathcal{B} = (e_1, \dots, e_n)$ be an orthonormal basis for V and $\mathcal{C} = (f_1, \dots, f_m)$ an orthonormal basis for W . Then

$$[T^*]_{\mathcal{B}}^{\mathcal{C}} = \left(\overline{[T]_{\mathcal{C}}^{\mathcal{B}}} \right)^T = \left([T]_{\mathcal{C}}^{\mathcal{B}} \right)^*,$$

where the last asterisk denotes the adjoint matrix.

Corollary 8.5

Let $A \in M_{m \times n}(K)$, where $K = \mathbb{R}$ or $\mathbb{C}$ and let $T_A : K^n \rightarrow K^m$ be the linear map $T_A(v) = Av$. Endow K^n and K^m with their standard inner product. Then

$$(T_A)^* = T_{A^*}.$$

Proposition 8.6: Properties of the Adjoint Matrix

Let $K = \mathbb{R}$ or $\mathbb{C}$, $A, B \in M_{n \times m}(K)$, $C \in M_{m \times p}(K)$. Then:

1. $(A + B)^* = A^* + B^*$.
2. $(\lambda A)^* = \bar{\lambda} A^* \quad \forall \lambda \in K$.
3. $(A^*)^* = A$.
4. $I_n^* = I_n$.
5. $(AC)^* = C^* A^*$.

9 Spectral Theorems

Definition 9.1: Orthogonal Diagonalizability

Let V be an inner product space and $T : V \rightarrow V$ an endomorphism. We say that T is **orthogonally diagonalizable** if there exists an orthonormal basis for V consisting of eigenvectors of T .

Lemma 9.2: Consequence of Orthogonal Diagonalizability

If T is orthogonally diagonalizable, then

$$T \circ T^* = T^* \circ T.$$

Definition 9.3: Normal Endomorphism/Matrix

Let V be an inner product space. An endomorphism $T : V \rightarrow V$ is called **normal** if

$$T \circ T^* = T^* \circ T.$$

A matrix $A \in M_{n \times n}(K)$ ($K = \mathbb{R}, \mathbb{C}$) is called **normal** if

$$AA^* = A^*A.$$

This is equivalent to saying that $T_A : K^n \rightarrow K^n$ is normal, if we endow K^n with its standard inner product.

We've seen in Lemma 9.2 that if $T : V \rightarrow V$ is orthogonally diagonalizable, then T is normal. Is the converse statement also true?

Over $K = \mathbb{R}$ the answer is **no!**, e.g.,

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

$T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a clockwise rotation by 90° , so there exist no eigenvalues in $\mathbb{R}$. But $AA^* = A^*A = I$.

Theorem 9.4: Spectral Theorem over $\mathbb{C}$

Let V be a finite dimensional inner product space over $\mathbb{C}$ and $T : V \rightarrow V$ a linear map. Then T is orthogonally diagonalizable if and only if T is normal.

Lemma 9.5: Properties of a Normal Endomorphism

Let V be a finite dimensional inner product space over $K = \mathbb{R}$ or $\mathbb{C}$. Let $T : V \rightarrow V$ be a normal endomorphism. Then:

1. $\forall v \in V : \|Tv\| = \|T^*v\|$.
2. $\forall \lambda \in K$ the endomorphism $T - \lambda \cdot \text{id}_V$ is also normal.
3. If v is an eigenvector of T with eigenvalue λ , then v is also an eigenvector of T^* with eigenvalue $\bar{\lambda}$.
4. Let v_1, v_2 be eigenvectors of T with eigenvalues λ_1, λ_2 respectively. If $\lambda_1 \neq \lambda_2$, then $\langle v_1, v_2 \rangle = 0$.

Proof. 1) Let $v \in V$. Then by the properties of the adjoint map T^* and since T is normal, we have that

$$\begin{aligned} \|Tv\|^2 &= \langle Tv, Tv \rangle = \langle v, T^* \circ Tv \rangle = \langle v, T \circ T^*v \rangle \\ &= \overline{\langle T \circ T^*v, v \rangle} = \overline{\langle T^*v, T^*v \rangle} = \langle T^*v, T^*v \rangle = \|T^*v\|^2. \end{aligned}$$

Since the norm is non-negative, we conclude that

$$\|Tv\| = \|T^*v\|.$$

2) Let $\lambda \in K$. By properties of adjoint maps and the normality of T we have that □